Nonlinear Dynamics and Chaos

Comprehensive Lecture Notes

Based on the course following
Nonlinear Dynamics and Chaos (3rd Edition)
by Steven H. Strogatz

Spring 2026

These notes are self-contained and assume familiarity with multivariable calculus, linear algebra, and ordinary differential equations at the level of a mathematical methods of physics course.

Contents

Overview: The Hierarchy of Dynamical Systems **5**

I One-Dimensional Flows **6**

1 Flows on the Line **6**
1.1 First-Order ODEs and Autonomous Systems . . . 6
1.2 A Geometric Way of Thinking: The Phase Line . . . 6
1.3 Linear Stability Analysis . . . 7
1.4 Existence and Uniqueness . . . 7
1.5 Impossibility of Oscillations . . . 7
1.6 Potentials . . . 8
1.7 Solving Equations on the Computer: Euler's Method . . . 8

2 Bifurcations **8**
2.1 Introduction . . . 8
2.2 Saddle-Node Bifurcation . . . 9
2.3 Transcritical Bifurcation . . . 9
2.4 Pitchfork Bifurcation . . . 10
2.4.1 Supercritical Pitchfork . . . 10
2.4.2 Subcritical Pitchfork . . . 10
2.5 Imperfect Bifurcations and Catastrophes . . . 10
2.6 Summary of 1D Bifurcations . . . 11
2.7 Application: Overdamped Bead on a Rotating Hoop . . . 11
2.8 Application: Insect Outbreak (Spruce Budworm) . . . 11

3 Flows on the Circle **12**
3.1 Motivation and Definitions . . . 12
3.2 Uniform and Nonuniform Oscillators . . . 12
3.3 The Overdamped Pendulum . . . 12
3.4 Superconducting Josephson Junctions . . . 12

II Two-Dimensional Flows **13**

4 Linear Systems in Two Dimensions **13**
4.1 Setup . . . 13
4.2 Solution by Eigenvalues and Eigenvectors . . . 13
4.3 Classification of Fixed Points . . . 13

5 Phase Plane Analysis **14**
5.1 Nonlinear Systems and Phase Portraits . . . 14
5.2 Existence, Uniqueness, and Topological Consequences . . . 15
5.3 Linearization Near Fixed Points . . . 15
5.4 Application: Competing Species (Rabbits vs. Sheep) . . . 15
5.5 Application: The SIR Epidemic Model . . . 16

5.6 Conservative Systems . . . 16
5.7 Reversible Systems . . . 17
5.8 Index Theory . . . 17

6 Limit Cycles **18**
6.1 Definitions and Examples . . . 18
6.2 Ruling Out Closed Orbits . . . 18
6.2.1 Bendixson's Negative Criterion . . . 18
6.2.2 Gradient Systems . . . 18
6.2.3 Liapunov Functions . . . 19
6.2.4 Dulac's Criterion . . . 19
6.3 Poincaré–Bendixson Theorem . . . 19
6.4 Liénard Systems . . . 20
6.5 The Van der Pol Oscillator . . . 20
6.6 Relaxation Oscillations . . . 20

7 Bifurcations Revisited: Two-Dimensional Systems **21**
7.1 Saddle-Node, Transcritical, and Pitchfork in 2D . . . 21
7.2 Hopf Bifurcation . . . 21
7.2.1 Supercritical Hopf Bifurcation . . . 21
7.2.2 Subcritical Hopf Bifurcation . . . 21
7.3 Global Bifurcations of Cycles . . . 22
7.4 Hysteresis in the Josephson Junction (2D) . . . 22
7.5 Coupled Oscillators and Quasiperiodicity . . . 22
7.5.1 Dynamics on the Torus . . . 23
7.6 Poincaré Maps . . . 23

8 Weakly Nonlinear Oscillations **23**
8.1 Motivation . . . 23
8.2 Two-Timing (Multiple Scale Analysis) . . . 23
8.3 Averaging Method . . . 24
8.4 Frequency Shift (Duffing Oscillator) . . . 24

III Chaos **24**

9 The Lorenz Equations **24**
9.1 Origin and Physical Motivation . . . 24
9.2 Physical Derivation: The Lorenz Waterwheel . . . 25
9.3 Simple Properties . . . 25
9.4 Chaos on the Strange Attractor . . . 26
9.5 The Lorenz Map . . . 26
9.6 Exploring Parameter Space of the Lorenz System . . . 27
9.7 The AC-Driven Pendulum and Routes to Chaos . . . 27

10 Lyapunov Exponents **27**
10.1 Definition for Continuous-Time Systems . . . 27
10.2 The Lyapunov Spectrum . . . 28
10.3 Lyapunov Exponents for the Lorenz System . . . 28
10.4 Kaplan–Yorke (Lyapunov) Dimension . . . 28

11 What Is Chaos? **29**

12 Routes to Chaos **29**
12.1 Route 1: Period-Doubling Cascade (Feigenbaum) . . . 29
12.2 Route 2: Quasiperiodic Route (Ruelle–Takens–Newhouse) . . . 30
12.3 Route 3: Intermittency (Pomeau–Manneville, 1980) . . . 31
12.4 Summary: Signatures of the Three Routes . . . 31

13 One-Dimensional Maps **31**
13.1 Introduction . . . 31
13.2 Fixed Points and Cobweb Diagrams . . . 32
13.3 Periodic Orbits . . . 32
13.4 The Logistic Map . . . 32
13.4.1 Fixed Points . . . 32
13.4.2 Period Doubling . . . 33
13.4.3 Period-Doubling Cascade and Feigenbaum's Universality . . . 33
13.4.4 Periodic Windows . . . 33
13.4.5 Invariant Measure of the Logistic Map at $r = 4$ . . . 34
13.5 Lyapunov Exponent . . . 34
13.6 The Schwarzian Derivative . . . 34
13.7 Renormalization . . . 35

14 Symbolic Dynamics and the Tent Map **35**
14.1 The Tent Map . . . 35
14.2 Symbolic Dynamics . . . 35
14.3 Topological Conjugacy . . . 36
14.4 Smale Horseshoe . . . 36

15 Fractals and Strange Attractors **37**
15.1 Fractal Dimension . . . 37
15.2 Dimension of Self-Similar Fractals . . . 37

16 The Rössler System **37**
16.1 Equations and Motivation . . . 37
16.2 Fixed Points . . . 38
16.3 The Rössler Attractor and Period-Doubling Route . . . 38
16.4 Comparison: Rössler vs. Lorenz . . . 38

17 The Hénon Map **39**
17.1 Definition and Motivation . . . 39
17.2 Properties . . . 39
17.3 Fixed Points and Stability . . . 39
17.4 The Hénon Strange Attractor . . . 39

17.5 Bifurcation Diagram of the Hénon Map . . . 40
17.6 Strange Attractors . . . 40
17.7 Correlation Dimension . . . 40
17.8 Crises: Sudden Changes in Chaotic Attractors . . . 41
17.9 Summary: Properties of Chaotic Attractors . . . 41

IV Collective Behavior 41

18 The Kuramoto Model 41
18.1 Motivation: Synchronization in Nature . . . 41
18.2 The Model . . . 42
18.3 The Order Parameter . . . 42
18.4 Self-Consistency Analysis (Infinite N) . . . 42
18.5 Critical Coupling and the Phase Transition . . . 43

V Additional Topics 43

19 Hamiltonian Systems and KAM Theory 43
19.1 Hamiltonian Systems . . . 43
19.2 Integrable Systems and Action-Angle Variables . . . 44
19.3 Chaos in Hamiltonian Systems . . . 44

20 Controlling Chaos 45
20.1 The OGY Method . . . 45

A Summary of Key Theorems 45

B Normal Forms and Center Manifold 46
B.1 Center Manifold Theorem . . . 46
B.2 Normal Forms . . . 46

C Useful Mathematical Facts 47
C.1 Taylor Expansion for Stability Analysis . . . 47
C.2 The Jacobian Matrix . . . 47
C.3 Green's Theorem (for Dulac's Criterion) . . . 47
C.4 Eigenvalues of a 2×2 Matrix . . . 47
C.5 Polar Coordinate Conversions . . . 47

Overview: The Hierarchy of Dynamical Systems

A **dynamical system** is any rule that describes how a state evolves in time. We study systems of the general form $\dot{\vec{x}} = \vec{F}(\vec{x})$ (continuous time) or $\vec{x}_{n+1} = \vec{F}(\vec{x}_n)$ (discrete time).

Hierarchy by dimension.

Dim.	Phase space	Possible attractors	Key tools
1	Real line $\mathbb{R}$	Fixed points only	Phase line, potentials
1	Circle S^1	Fixed pts, oscillations	SNIPER bifurcation
2	Plane $\mathbb{R}^2$	Fixed pts, limit cycles	Poincaré–Bendixson
3+	$\mathbb{R}^n$	Above + strange attractors	Lyapunov exponents

Linear vs. nonlinear. A linear system $\dot{\vec{x}} = A\vec{x}$ can be solved exactly by exponentiation: $\vec{x}(t) = e^{At}\vec{x}_0$. It has no limit cycles or chaos. Nonlinearity is *necessary* for the rich behavior studied in this course: bifurcations, limit cycles, strange attractors.

Continuous (flows) vs. discrete (maps). Maps arise naturally as Poincaré sections of flows, lowering the dimension by one. Chaos is possible in 1D maps but requires at least 3D in continuous-time autonomous flows (Poincaré–Bendixson theorem).

Key phenomena covered in these notes.

1. **Fixed points and their stability** (everywhere).
2. **Bifurcations**: qualitative changes as a parameter crosses a critical value.
3. **Oscillations**: limit cycles, flows on the circle.
4. **Chaos**: sensitive dependence, strange attractors, Lyapunov exponents, fractal geometry.
5. **Collective behavior**: synchronization (Kuramoto model).

Part I
One-Dimensional Flows

1 Flows on the Line

1.1 First-Order ODEs and Autonomous Systems

The most general first-order ordinary differential equation (ODE) for a single real variable $x(t)$ is

$$\dot{x} = f(x, t), \tag{1}$$

where the overdot denotes $\mathrm{d}x \,/\, \mathrm{d}t$.

Definition 1.1 (Autonomous ODE)**.** An ODE of the form

$$\dot{x} = f(x) \tag{2}$$

is called **autonomous** because the right-hand side does not depend explicitly on time t. The function f specifies the *velocity* of the state x on the real line.

The system (2) is **linear** if $f(x) = ax + b$ for constants a, b, and **nonlinear** otherwise.

Given an initial condition $x(0) = x_0$, the solution $x(t)$ traces a trajectory on the real line. Because f depends only on x, the velocity at a given position x is the same regardless of when the particle passes through x. This is the hallmark of an autonomous system.

1.2 A Geometric Way of Thinking: The Phase Line

Rather than solving (2) analytically (which is often impossible for nonlinear f), we adopt a *geometric* approach.

Procedure.

1. Plot $f(x)$ versus x.
2. Where $f(x) > 0$, the flow moves to the *right* (increasing x); where $f(x) < 0$, the flow moves to the *left*.
3. Points where $f(x^*) = 0$ are at rest—they are called **fixed points** (or equilibria, critical points, stationary points).

Definition 1.2 (Fixed Point)**.** A point x^* is a **fixed point** of $\dot{x} = f(x)$ if $f(x^*) = 0$.

Definition 1.3 (Stability of a Fixed Point—Informal)**.** A fixed point x^* is

- **stable** (an "attractor") if nearby trajectories approach x^* as $t \to \infty$;
- **unstable** (a "repeller") if nearby trajectories move away from x^*.

Graphical rule. Looking at the graph of $f(x)$: a fixed point where f crosses from positive to negative (i.e., $f'(x^*) < 0$) is stable; a crossing from negative to positive ($f'(x^*) > 0$) is unstable.

Direction Field

For autonomous systems, the direction field has identical slope $f(x)$ at every point along a horizontal line $x = \text{const}$. Solution trajectories in the (t, x) plane are parallel translations of each other. The initial condition x_0 simply selects which trajectory the system follows.

1.3 Linear Stability Analysis

To make the stability criterion precise, we linearize near the fixed point.

Let x^* be a fixed point and set $u = x - x^*$ (a small perturbation). Then

$$\dot{u} = \dot{x} = f(x) = f(x^*) + f'(x^*)\,u + O(u^2).$$

Since $f(x^*) = 0$, dropping higher-order terms gives the *linearized equation*:

$$\dot{u} \approx f'(x^*)\,u. \tag{3}$$

This is a linear ODE with solution

$$u(t) = u_0\, e^{f'(x^*)\,t}. \tag{4}$$

Linear Stability Criterion (1D)

Let x^* be a fixed point of $\dot{x} = f(x)$.

- If $f'(x^*) < 0$, then x^* is **linearly stable**.
- If $f'(x^*) > 0$, then x^* is **unstable**.
- If $f'(x^*) = 0$, the test is **inconclusive**—higher-order terms must be examined.

Example 1.1 (Population Growth)**.** Consider $\dot{x} = rx(1 - x/K)$ (the logistic equation) with $r, K > 0$. The fixed points are $x^* = 0$ and $x^* = K$. Since $f'(x) = r - 2rx/K$, we have $f'(0) = r > 0$ (unstable) and $f'(K) = -r < 0$ (stable). The carrying capacity K is the stable equilibrium.

1.4 Existence and Uniqueness

Theorem 1.1 (Existence and Uniqueness)**.** *Consider the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$. If $f(x)$ is continuous and $f'(x)$ is also continuous (i.e., $f \in C^1$) on an open interval containing x_0, then a solution $x(t)$ exists and is unique on some time interval $(-\tau, \tau)$ about $t = 0$.*

Remark 1.1. A key consequence is that trajectories on the real line *cannot cross* fixed points. This means the qualitative behavior is completely determined by the signs of $f(x)$ between consecutive fixed points.

1.5 Impossibility of Oscillations

Theorem 1.2. *For a first-order autonomous ODE $\dot{x} = f(x)$ on the real line with $f \in C^1$, oscillations (periodic solutions) are impossible.*

Proof. The velocity $f(x)$ at any point x is single-valued and time-independent. For the system to oscillate, x would have to return to a previously visited value, requiring $\dot{x}$ to change sign while passing through the same point—but $f(x)$ is fixed at each x. More precisely, by the uniqueness theorem, trajectories cannot cross, and on $\mathbb{R}^1$ there is no way to return to a starting point without crossing a fixed point, at which the solution would be trapped forever. □

1.6 Potentials

For one-dimensional flows, we can often think of the dynamics in terms of a "particle rolling downhill on a potential landscape."

Definition 1.4 (Potential Function)**.** Given $\dot{x} = f(x)$, define the **potential** $V(x)$ by

$$f(x) = -\frac{\mathrm{d}V}{\mathrm{d}x}, \qquad \text{i.e.} \qquad V(x) = -\int f(x)\,dx. \tag{5}$$

Then $\dot{x} = -V'(x)$: the system always moves in the direction of decreasing V. Stable fixed points correspond to local *minima* of V; unstable fixed points correspond to local *maxima*.

The potential decreases monotonically along trajectories:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = V'(x)\,\dot{x} = V'(x)\big[-V'(x)\big] = -\big[V'(x)\big]^2 \leq 0.$$

1.7 Solving Equations on the Computer: Euler's Method

In practice, many nonlinear ODEs cannot be solved in closed form. The simplest numerical scheme for $\dot{x} = f(x)$ is **Euler's method**:

$$x_{n+1} = x_n + f(x_n)\,\Delta t, \tag{6}$$

which approximates the solution by taking small steps of size Δt along the tangent line. More sophisticated methods (Runge–Kutta, adaptive step-size) are used in practice, but Euler's method captures the essential idea.

Example 1.2 (Separation of Variables and Finite-Time Blowup)**.** Consider $\dot{x} = 1 + x^2$ with $x(0) = 0$. By separation of variables:

$$\int \frac{dx}{1+x^2} = \int dt \quad \Longrightarrow \quad \arctan(x) = t + C.$$

With $x(0) = 0$: $C = 0$, so $x(t) = \tan(t)$. The solution blows up in finite time $t = \pi/2$.

This illustrates an important caveat: the existence and uniqueness theorem guarantees solutions only *locally* in time. If f grows fast enough, the solution can escape to infinity in finite time. However, for flows on the *line* where f is smooth and bounded, this cannot happen.

Example 1.3 (Exact Solution: $\dot{x} = x^2 - 1$)**.** Fixed points: $x^* = \pm 1$. Since $f'(x) = 2x$, $f'(-1) = -2 < 0$ (stable) and $f'(1) = 2 > 0$ (unstable). By partial fractions:

$$\int \frac{dx}{x^2-1} = \int \frac{dx}{(x-1)(x+1)} = \frac{1}{2}\int \left(\frac{1}{x-1} - \frac{1}{x+1}\right) dx = t + C.$$

Solving gives $x(t) = -\tanh(t - t_0)$ for trajectories between the fixed points, confirming the graphical analysis.

2 Bifurcations

2.1 Introduction

A **bifurcation** occurs when a small, smooth change in a parameter causes a qualitative (topological) change in the dynamics—for instance, fixed points may be created, destroyed, or change stability.

The parameter is traditionally denoted r. The key question is: *How does the phase portrait of $\dot{x} = f(x; r)$ change as r varies?*

A **bifurcation diagram** plots a characteristic feature of the solution (usually the fixed point values x^*) on the vertical axis versus the parameter r on the horizontal axis. Solid curves denote stable fixed points; dashed curves denote unstable ones.

2.2 Saddle-Node Bifurcation

The **saddle-node bifurcation** is the basic mechanism by which fixed points are created and destroyed.

Normal Form: Saddle-Node Bifurcation

$$\dot{x} = r + x^2. \tag{7}$$

Analysis.

- For $r < 0$: two fixed points at $x^* = \pm\sqrt{-r}$. Since $f'(x) = 2x$, the negative root is stable ($f' < 0$) and the positive root is unstable ($f' > 0$).
- At $r = 0$: the two fixed points coalesce into one *half-stable* fixed point at $x^* = 0$.
- For $r > 0$: no fixed points exist—$f(x) > 0$ everywhere so x increases without bound.

The bifurcation occurs at $r_c = 0$. Graphically, the two branches of the bifurcation diagram meet in a cusp.

Bottleneck / Ghost Effect

Just above the bifurcation ($r = 0^+$), even though no fixed point exists, the system slows dramatically near $x = 0$ because $f(x) \approx r$ is small. The time to pass through this "bottleneck" scales as $T_{\text{bottleneck}} \sim \pi/\sqrt{r}$ for $r \to 0^+$. This is sometimes called the *ghost* of the vanished fixed point.

2.3 Transcritical Bifurcation

In many physical situations, a fixed point exists for all parameter values but exchanges stability with another fixed point.

Normal Form: Transcritical Bifurcation

$$\dot{x} = rx - x^2 = x(r - x). \tag{8}$$

Analysis. Fixed points: $x^* = 0$ and $x^* = r$. Since $f'(x) = r - 2x$:

- $f'(0) = r$: origin is stable for $r < 0$, unstable for $r > 0$.
- $f'(r) = -r$: the fixed point $x^* = r$ is unstable for $r < 0$, stable for $r > 0$.

At $r = 0$ the two fixed points collide and **exchange stability**. Neither fixed point is destroyed; they simply swap roles.

Laser Threshold

A laser below threshold ($r < 0$) has zero output (stable $x^* = 0$). Above threshold ($r > 0$), $x^* = 0$ becomes unstable and a new stable state $x^* = r$ (proportional to the laser output intensity) takes over. This is a transcritical bifurcation.

2.4 Pitchfork Bifurcation

Pitchfork bifurcations arise in systems that possess a **symmetry** $x \to -x$. There are two types.

2.4.1 Supercritical Pitchfork

Normal Form: Supercritical Pitchfork

$$\dot{x} = rx - x^3. \tag{9}$$

Analysis. Fixed points: $x^* = 0$ for all r, and $x^* = \pm\sqrt{r}$ for $r > 0$.

Since $f'(x) = r - 3x^2$:

- $f'(0) = r$: origin is stable for $r < 0$, unstable for $r > 0$.
- $f'(\pm\sqrt{r}) = r - 3r = -2r < 0$ for $r > 0$: the two new branches are stable.

At $r = 0$, the origin loses stability and two new stable branches are born symmetrically. The bifurcation diagram looks like a "pitchfork."

2.4.2 Subcritical Pitchfork

Normal Form: Subcritical Pitchfork

$$\dot{x} = rx + x^3. \tag{10}$$

Here the cubic term is *destabilizing*. For $r < 0$, there are three fixed points: the origin (stable) and $x^* = \pm\sqrt{-r}$ (both unstable). At $r = 0$ the unstable branches merge with the origin, making it unstable for $r > 0$.

Subcritical Pitchfork and Hysteresis

In a physical system, the subcritical pitchfork is often "stabilized" by a higher-order term, e.g., $\dot{x} = rx + x^3 - x^5$. This leads to **hysteresis**: the system can jump discontinuously between branches as r is varied, and the value of x observed depends on the *direction* in which r is changed.

2.5 Imperfect Bifurcations and Catastrophes

When the symmetry of a pitchfork bifurcation is slightly broken (an "imperfection" h), the bifurcation diagram deforms. The normal form is

$$\dot{x} = h + rx - x^3. \tag{11}$$

For $h \neq 0$, the perfect pitchfork breaks into an isolated branch and an S-shaped curve. The set of bifurcation points in the (r, h) parameter plane forms a cusp—this is the **cusp catastrophe**, one of the elementary catastrophes of singularity theory.

2.6 Summary of 1D Bifurcations

Bifurcation	**Normal Form**	**Mechanism**
Saddle-node	$\dot{x} = r + x^2$	Two FPs created/destroyed
Transcritical	$\dot{x} = rx - x^2$	FPs exchange stability
Supercrit. pitchfork	$\dot{x} = rx - x^3$	One FP $\to$ three; stable branches
Subcrit. pitchfork	$\dot{x} = rx + x^3$	Three FPs $\to$ one; unstable branches

2.7 Application: Overdamped Bead on a Rotating Hoop

A bead slides without friction on a circular wire hoop of radius R that rotates about its vertical diameter with angular velocity ω. In the overdamped limit (viscous damping dominates inertia), the angle ϕ of the bead from the bottom of the hoop obeys:

$$\dot{\phi} = f(\phi) = \sin\phi\big(\gamma\cos\phi - 1\big), \tag{12}$$

where $\gamma = R\omega^2/g$ is the dimensionless rotation rate.

Analysis:

- Fixed points: $\phi^* = 0, \pi$ always exist. For $\gamma > 1$, two additional fixed points appear at $\phi^* = \pm\arccos(1/\gamma)$.
- At $\gamma = 1$: a **supercritical pitchfork bifurcation** occurs. The bottom ($\phi = 0$) changes from stable to unstable, and the bead spontaneously climbs to one of the new off-axis equilibria.

This is a beautiful physical realization of symmetry-breaking: for slow rotation the bead sits at the bottom; for fast rotation it "chooses" a side, breaking the left-right symmetry.

2.8 Application: Insect Outbreak (Spruce Budworm)

The budworm population N satisfies a model of the form

$$\dot{N} = rN\left(1 - \frac{N}{K}\right) - p(N), \tag{13}$$

where the first term is logistic growth and $p(N) = BN^2/(A^2 + N^2)$ is predation (a sigmoidal functional response). Nondimensionalizing with $x = N/A$, $\tau = rAt$, etc., leads to

$$\dot{x} = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}. \tag{14}$$

For appropriate parameter ranges, this system exhibits **hysteresis** via saddle-node bifurcations. As parameters change slowly (e.g., the forest matures, increasing k), the budworm population can undergo a sudden catastrophic outbreak (a jump to a high-x branch), followed by an equally sudden crash when the branch disappears. This involves two saddle-node bifurcations forming a hysteresis loop—an example of a **cusp catastrophe** in an ecological setting.

3 Flows on the Circle

3.1 Motivation and Definitions

On the real line, oscillations are impossible for a first-order autonomous system. However, if the phase space is a **circle** S^1 (i.e., the variable θ is 2π-periodic), oscillations *are* possible: the system can flow around the circle indefinitely.

Definition 3.1. A **flow on the circle** is an ODE of the form

$$\dot{\theta} = f(\theta), \qquad \theta \in [0, 2\pi), \tag{15}$$

where $f(\theta + 2\pi) = f(\theta)$.

Fixed points are again defined by $f(\theta^*) = 0$, and stability analysis is identical to the real-line case.

3.2 Uniform and Nonuniform Oscillators

Uniform oscillator: $\dot{\theta} = \omega$ (constant angular velocity). Period $T = 2\pi/\omega$.

Nonuniform oscillator: $\dot{\theta} = \omega - a\sin\theta$ with $a, \omega > 0$.

- If $\omega > a$: $\dot{\theta} > 0$ always, so θ oscillates but nonuniformly. The period is

$$T = \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \tag{16}$$

- At $\omega = a$: a saddle-node bifurcation creates a half-stable fixed point. The period diverges: $T \to \infty$.
- For $\omega < a$: two fixed points appear (one stable, one unstable) and oscillation ceases.

This is the prototypical example of a **saddle-node bifurcation on a circle**, also called a SNIPER (Saddle-Node Infinite-PERiod) bifurcation. The divergence of the period scales as $T \sim 1/\sqrt{\omega - a}$ near the bifurcation.

3.3 The Overdamped Pendulum

A pendulum in a viscous fluid with a constant applied torque Γ is modeled by

$$\dot{\theta} = \Gamma - \sin\theta. \tag{17}$$

This is exactly the nonuniform oscillator with $\omega = \Gamma$ and $a = 1$. For $|\Gamma| < 1$, the pendulum settles to a tilted equilibrium; for $|\Gamma| > 1$, it whirls continuously.

3.4 Superconducting Josephson Junctions

A Josephson junction consists of two superconductors separated by a thin insulating barrier. The *phase difference* ϕ across the junction obeys (in the overdamped, zero-capacitance limit):

$$\frac{\hbar}{2eR}\,\dot{\phi} = I - I_c \sin\phi, \tag{18}$$

where I is the applied (DC) current, I_c is the critical current, and R is the shunt resistance. Nondimensionalizing:

$$\dot{\phi} = i - \sin\phi, \qquad i \equiv I/I_c. \tag{19}$$

This is a flow on the circle identical to the overdamped pendulum.

- For $|i| < 1$: two fixed points exist; $\dot{\phi} = 0$ at steady state and the voltage $V = (\hbar/2e)\,\dot{\phi} = 0$. The junction carries a **supercurrent** with zero resistance.
- For $|i| > 1$: no fixed points; ϕ winds continuously and a nonzero DC voltage appears across the junction.

Part II
Two-Dimensional Flows

4 Linear Systems in Two Dimensions

4.1 Setup

We now consider systems of two coupled first-order ODEs:

$$\dot{\vec{x}} = A\vec{x}, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}. \tag{20}$$

4.2 Solution by Eigenvalues and Eigenvectors

We seek solutions of the form $\vec{x}(t) = e^{\lambda t}\vec{v}$. Substituting into (20):

$$\lambda e^{\lambda t}\vec{v} = A\,e^{\lambda t}\vec{v} \quad \Longrightarrow \quad A\vec{v} = \lambda\vec{v}.$$

Thus λ must be an **eigenvalue** of A and $\vec{v}$ the corresponding **eigenvector**.

The eigenvalues satisfy the characteristic equation:

$$\lambda^2 - \tau\lambda + \Delta = 0, \tag{21}$$

where $\tau \equiv \operatorname{tr} A = a + d$ and $\Delta \equiv \det A = ad - bc$. The solutions are:

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}. \tag{22}$$

The general solution is $\vec{x}(t) = c_1 e^{\lambda_1 t}\vec{v}_1 + c_2 e^{\lambda_2 t}\vec{v}_2$ (when the eigenvalues are distinct).

4.3 Classification of Fixed Points

The qualitative behavior of trajectories near the origin depends on the eigenvalues, which are determined by τ and Δ.

Classification in the (τ, Δ) Plane

Region	Eigenvalues	Fixed Point Type
$\Delta < 0$	Real, opposite signs	**Saddle point**
$\Delta > 0$, $\tau^2 > 4\Delta$, $\tau < 0$	Real, both negative	**Stable node**
$\Delta > 0$, $\tau^2 > 4\Delta$, $\tau > 0$	Real, both positive	**Unstable node**
$\Delta > 0$, $\tau^2 < 4\Delta$, $\tau < 0$	Complex, Re < 0	**Stable spiral**
$\Delta > 0$, $\tau^2 < 4\Delta$, $\tau > 0$	Complex, Re > 0	**Unstable spiral**
$\Delta > 0$, $\tau = 0$	Pure imaginary	**Center** (linear)
$\Delta > 0$, $\tau^2 = 4\Delta$	Real, repeated	**Star node** or **degenerate node**
$\Delta = 0$	One eigenvalue zero	**Line of fixed points**

Key relationships:

$$\lambda_1 + \lambda_2 = \tau = \operatorname{tr} A, \tag{23}$$

$$\lambda_1 \cdot \lambda_2 = \Delta = \det A. \tag{24}$$

The stability boundary is $\tau = 0$ (eigenvalues cross the imaginary axis). The boundary between nodes and spirals is $\tau^2 = 4\Delta$. Saddle points live below the $\Delta = 0$ axis.

Physical Interpretation

- **Stable node:** trajectories approach the origin along straight lines (the eigenvector directions); overdamped approach.
- **Stable spiral:** trajectories spiral inward; underdamped oscillatory decay.
- **Center:** closed elliptical orbits; no damping, no growth.
- **Saddle:** trajectories approach along one eigenvector direction and are repelled along the other.

5 Phase Plane Analysis

5.1 Nonlinear Systems and Phase Portraits

We now consider the general two-dimensional nonlinear system:

$$\dot{x} = f(x, y), \qquad \dot{y} = g(x, y). \tag{25}$$

The **phase portrait** is the collection of all trajectories in the (x, y) plane.

Definition 5.1 (Nullclines)**.** The **x-nullcline** is the set of points where $\dot{x} = 0$, i.e., $f(x, y) = 0$. The **y-nullcline** is the set where $\dot{y} = 0$, i.e., $g(x, y) = 0$. Fixed points lie at the *intersections* of the two nullclines.

Nullclines are extremely useful for sketching phase portraits: on the x-nullcline, the flow is purely vertical ($\dot{x} = 0$); on the y-nullcline, the flow is purely horizontal ($\dot{y} = 0$).

5.2 Existence, Uniqueness, and Topological Consequences

Theorem 5.1 (Existence and Uniqueness in 2D)**.** *If f and g are continuously differentiable, then given any initial condition (x_0, y_0), a unique solution exists (at least locally in time).*

Corollary 5.2. *Different trajectories can never cross in the phase plane.*

This has profound consequences: for example, a closed orbit divides the plane into an "inside" and an "outside," and no trajectory can cross it.

5.3 Linearization Near Fixed Points

Let (x^*, y^*) be a fixed point of (25). Set $u = x - x^*$, $v = y - y^*$. Taylor expanding:

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = J\Big|_{(x^*,y^*)} \begin{pmatrix} u \\ v \end{pmatrix} + \text{h.o.t.}, \tag{26}$$

where J is the **Jacobian matrix**:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}\Bigg|_{(x^*,y^*)}. \tag{27}$$

Theorem 5.3 (Hartman–Grobman Theorem—Informal)**.** *If the fixed point is* ***hyperbolic*** *(i.e., $Re(\lambda_i) \neq 0$ for all eigenvalues of J), then the phase portrait of the nonlinear system near the fixed point is* qualitatively the same *as the phase portrait of the linearized system $\dot{\vec{u}} = J\vec{u}$.*

Borderline Cases

- **Centers** ($\tau = 0$, $\Delta > 0$) are *not* hyperbolic. A linear center may become a spiral (stable or unstable) in the nonlinear system. Additional analysis (e.g., conserved quantities, reversibility) is needed.
- **Non-isolated fixed points** ($\Delta = 0$) also require special treatment.

5.4 Application: Competing Species (Rabbits vs. Sheep)

Consider the Lotka–Volterra competition model:

$$\dot{x} = x(3 - x - 2y), \qquad \dot{y} = y(2 - x - y), \tag{28}$$

where x and y represent two competing populations.

Nullclines: $\dot{x} = 0$: $x = 0$ or $x + 2y = 3$. $\dot{y} = 0$: $y = 0$ or $x + y = 2$.

Fixed points: $(0, 0)$, $(3, 0)$, $(0, 2)$, and the intersection of $x + 2y = 3$ and $x + y = 2$, giving $(1, 1)$.

Linearizing at each fixed point using the Jacobian:

- $(0, 0)$: $J = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, eigenvalues $3, 2 > 0$: unstable node.
- $(3, 0)$: $J = \left(\begin{smallmatrix} -3 & -6 \\ 0 & -1 \end{smallmatrix}\right)$, eigenvalues $-3, -1$: stable node (sheep extinct).
- $(0, 2)$: $J = \left(\begin{smallmatrix} -1 & 0 \\ -2 & -2 \end{smallmatrix}\right)$, eigenvalues $-1, -2$: stable node (rabbits extinct).
- $(1, 1)$: $\tau = -3$, $\Delta = 1$, $\tau^2 - 4\Delta = 5 > 0$: a saddle point (coexistence equilibrium is unstable).

The separatrix of the saddle at $(1, 1)$ divides the first quadrant into two basins of attraction: one leading to $(3, 0)$ (species x wins) and the other to $(0, 2)$ (species y wins). The outcome depends on initial conditions—**competitive exclusion**.

5.5 Application: The SIR Epidemic Model

The **SIR model** divides a fixed population into Susceptible (S), Infected (I), and Recovered (R) compartments, with $S + I + R = N$:

$$\dot{S} = -\beta SI, \qquad \dot{I} = \beta SI - \gamma I, \qquad \dot{R} = \gamma I. \tag{29}$$

Since $R = N - S - I$, we study the 2D system in (S, I):

$$\dot{S} = -\beta SI, \qquad \dot{I} = I(\beta S - \gamma). \tag{30}$$

Fixed points: The $I = 0$ axis (disease-free) is a line of equilibria. The interior has no fixed points with $I > 0$.

Nullclines: $\dot{S} = 0$ on $S = 0$ or $I = 0$; $\dot{I} = 0$ on $I = 0$ or $S = \gamma/\beta \equiv S_c$ (the **herd immunity threshold**).

Key result. Define the **basic reproduction number** $\mathcal{R}_0 = \beta N/\gamma$:

- If $S_0 < S_c$ (i.e., $\mathcal{R}_0 < 1$ effectively): $\dot{I} < 0$ always; epidemic dies out.
- If $S_0 > S_c$ (i.e., $\mathcal{R}_0 > 1$): epidemic peaks when $S = S_c$, then I decreases. The epidemic ends not because $I \to 0$ at S_c, but because S is depleted below S_c.

The system is not a fixed-point problem but a **trajectory problem**: solutions are curves in the (S, I) plane, flowing left (since $\dot{S} < 0$) and forming an arch. There are *no* closed orbits (Bendixson: $\nabla \cdot \vec{F} = -\beta I - (\beta S - \gamma)$, which can change sign, so we use a Liapunov function instead).

5.6 Conservative Systems

Definition 5.2 (Conservative System)**.** A system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ is **conservative** if there exists a function $E(x, y)$ (the *energy* or *conserved quantity*) such that E is constant along trajectories: $\frac{\mathrm{d}E}{\mathrm{d}t} = 0$.

Theorem 5.4. *Conservative systems have* ***no attractors****. Trajectories lie on level sets of E, and there can be no stable nodes, stable spirals, or stable limit cycles.*

Proof. Since E is constant along trajectories, no trajectory can approach a single point (which would require E to take a single value over an open neighborhood), nor can a trajectory spiral inward (which would require E to decrease monotonically). □

Example 5.1 (Harmonic Oscillator)**.** $m\ddot{x} = -kx$ can be written as $\dot{x} = v$, $\dot{v} = -(k/m)\,x$. The energy $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$ is conserved:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = kx\dot{x} + mv\dot{v} = kxv + mv\big(-\tfrac{k}{m}x\big) = 0.$$

The Jacobian at the origin has $\tau = 0$, $\Delta = k/m > 0$: a linear center. Since the system is conservative, this center persists as a true nonlinear center with closed elliptical orbits.

5.7 Reversible Systems

Definition 5.3 (Time-Reversibility)**.** A system is **reversible** if it is invariant under the transformation $t \to -t$ and some transformation of the state variables (typically $y \to -y$, or equivalently, the velocity reverses).

Theorem 5.5. *For a reversible system, if a trajectory lies on the line of symmetry (the axis that is invariant under the reversal), then any linear center is a* true *nonlinear center.*

Example 5.2 (Undamped Pendulum)**.** The equation $\ddot{\theta} = -\sin\theta$ written as a system: $\dot{\theta} = v$, $\dot{v} = -\sin\theta$. Under $t \to -t$, $v \to -v$: the equations are unchanged. Therefore the linear center at $(\theta, v) = (0, 0)$ is a true nonlinear center.

The energy is $E = \frac{1}{2}v^2 - \cos\theta$. Phase portrait:

- $E < 1$: librations (closed orbits around the center).
- $E = 1$: the **separatrix** (homoclinic orbits connecting the saddle at $(\pi, 0)$ to itself).
- $E > 1$: rotations (unbounded trajectories wrapping around).

5.8 Index Theory

Index theory provides topological constraints on the arrangement of fixed points and closed orbits.

Definition 5.4 (Index of a Closed Curve)**.** Let C be a simple closed curve in the phase plane that does not pass through any fixed points. At each point on C, the vector field $(\dot{x}, \dot{y})$ makes an angle ϕ with the positive x-axis. As we traverse C counterclockwise, ϕ changes continuously. The **index** of C is

$$I_C = \frac{1}{2\pi} \oint_C d\phi = \frac{[\phi]_C}{2\pi}, \tag{31}$$

i.e., the number of complete counterclockwise revolutions of the vector field as we traverse C once counterclockwise.

Theorem 5.6 (Properties of the Index)**.**

1. *If C encloses no fixed points, then $I_C = 0$.*
2. *If C is a trajectory (e.g., a closed orbit), then $I_C = +1$.*
3. *The index is unchanged if C is continuously deformed, as long as it does not cross a fixed point.*
4. *The index of a curve enclosing multiple isolated fixed points equals the sum of the indices of the individual fixed points.*
5. *Reversing time ($t \to -t$) does not change the index.*

Indices of Common Fixed Points

Fixed Point Type	Index
Stable/unstable node	$+1$
Stable/unstable spiral	$+1$
Center	$+1$
Saddle point	-1

Corollary 5.7. *Any closed orbit must enclose fixed points whose indices sum to $+1$. In particular, a closed orbit must enclose at least one fixed point that is not a saddle.*

6 Limit Cycles

6.1 Definitions and Examples

Definition 6.1 (Limit Cycle)**.** A **limit cycle** is an isolated closed orbit. "Isolated" means that neighboring trajectories are not closed—they either spiral toward or away from the limit cycle.

- **Stable** limit cycle: all nearby trajectories approach it as $t \to \infty$.
- **Unstable** limit cycle: all nearby trajectories move away.
- **Half-stable**: trajectories approach from one side and recede from the other.

Remark 6.1. Limit cycles are inherently **nonlinear** phenomena. They cannot occur in linear systems (where closed orbits, if they exist, form continuous families—centers—not isolated cycles).

Example 6.1 (Limit Cycle in Polar Coordinates)**.** Consider $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. The radial equation has fixed points $r^* = 0$ (unstable, since $f'(0) = 1 > 0$) and $r^* = 1$ (stable, since $f'(1) = -2 < 0$). Therefore $r = 1$ is a **stable limit cycle**: all trajectories with $r > 0$ spiral toward the unit circle.

6.2 Ruling Out Closed Orbits

Before searching for limit cycles, it is often useful to rule them out in certain regions.

6.2.1 Bendixson's Negative Criterion

Theorem 6.1 (Bendixson's Criterion)**.** *If $\nabla \cdot \vec{F} = \partial f/\partial x + \partial g/\partial y$ is not identically zero and does not change sign in a simply connected region R, then there are no closed orbits entirely contained in R.*

Proof. Suppose a closed orbit C exists in R. By Green's theorem:

$$\oint_C \vec{F} \cdot \hat{n}\, dl = \iint_A \nabla \cdot \vec{F}\, dA.$$

Since C is a trajectory, $\vec{F}$ is tangent to C everywhere, so $\vec{F} \cdot \hat{n} = 0$ and the left side vanishes. But if $\nabla \cdot \vec{F}$ has constant sign and is not identically zero, the right side is nonzero—a contradiction. □

Remark 6.2. Bendixson's criterion is a special case of Dulac's criterion with $g = 1$. It rules out limit cycles in systems where the divergence is one-signed. For example, the system $\dot{x} = y$, $\dot{y} = -x + y^2$ has $\nabla \cdot \vec{F} = 1 \neq 0$ (but constant), ruling out closed orbits.

6.2.2 Gradient Systems

If $\dot{x} = -\nabla V$ for some potential $V(x, y)$, then

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \nabla V \cdot \dot{\vec{x}} = \nabla V \cdot (-\nabla V) = -|\nabla V|^2 \leq 0.$$

So V decreases monotonically along trajectories, and closed orbits are impossible.

6.2.3 Liapunov Functions

Theorem 6.2 (Liapunov's Theorem for Ruling Out Closed Orbits)**.** *If there exists a function $V(x, y)$ satisfying:*

1. $V(x^*, y^*) = 0$ *at all fixed points,*
2. $V(x, y) > 0$ *for all (x, y) not a fixed point (in some region R),*
3. $\dot{V} = \frac{\mathrm{d}V}{\mathrm{d}t} < 0$ *for all non-fixed points in R,*

then there are no closed orbits entirely contained in R.

Proof sketch. On a closed orbit, V would have to return to its starting value after one period. But $\dot{V} < 0$ means V is strictly decreasing, a contradiction. □

6.2.4 Dulac's Criterion

Theorem 6.3 (Dulac's Criterion)**.** *Let R be a simply connected region of the plane. If there exists a continuously differentiable function $g(x, y)$ such that $\nabla \cdot (g\vec{F})$ does not change sign in R (and is not identically zero on any open subset), then there are no closed orbits entirely contained in R.*

Here $\vec{F} = (f, g_{sys})$ is the vector field, and we require

$$\frac{\partial(g\,f)}{\partial x} + \frac{\partial(g\,g_{sys})}{\partial y} \neq 0 \quad \textit{(same sign throughout } R\textit{).} \tag{32}$$

Proof. By Green's theorem, for any closed curve C bounding a region $A \subset R$:

$$\iint_A \nabla \cdot (g\vec{F})\, dA = \oint_C g\vec{F} \cdot \hat{n}\, dl.$$

If C were a closed orbit, then $\vec{F}$ is tangent to C everywhere, so $\vec{F} \cdot \hat{n} = 0$ and the right side vanishes. But the left side is nonzero by hypothesis, a contradiction. □

6.3 Poincaré–Bendixson Theorem

This theorem is one of the most powerful tools for *proving the existence* of limit cycles.

Theorem 6.4 (Poincaré–Bendixson Theorem)**.** *Suppose that:*

1. *R is a closed, bounded region of the plane;*
2. *$\dot{\vec{x}} = \vec{F}(\vec{x})$ is a continuously differentiable vector field on an open set containing R;*
3. *R does not contain any fixed points;*
4. *There exists a trajectory C that is "confined" to R (i.e., C starts in R and can never leave).*

Then C is either a closed orbit, or it spirals toward a closed orbit contained in R. In either case, R contains a ***closed orbit****.*

Remark 6.3. This theorem is specific to two dimensions. In three or more dimensions, a bounded trajectory need not approach a closed orbit (it could be a strange attractor, for example).

Trapping Regions

To apply the Poincaré–Bendixson theorem, one constructs a **trapping region**: a closed, bounded region R with no fixed points inside, such that the vector field points inward (or is tangent) on the entire boundary ∂R. Then any trajectory entering R is confined to it, and the theorem guarantees a closed orbit inside.
The boundary of R is often built from segments of nullclines, coordinate axes, or level curves of auxiliary functions.

6.4 Liénard Systems

Definition 6.2 (Liénard System)**.** A **Liénard system** is a second-order ODE of the form

$$\ddot{x} + f(x)\dot{x} + g(x) = 0, \tag{33}$$

or equivalently, the system $\dot{x} = y - F(x)$, $\dot{y} = -g(x)$, where $F(x) = \int_0^x f(s)\,ds$.

Theorem 6.5 (Liénard's Theorem)**.** *The Liénard system has a unique, stable limit cycle if:*

1. *f and g are continuous, and g is Lipschitz;*
2. *g is an odd function: $g(-x) = -g(x)$, with $g(x) > 0$ for $x > 0$;*
3. *F is an odd function with exactly one positive zero at $x = a$, $F(x) < 0$ for $0 < x < a$, $F(x) > 0$ and nondecreasing for $x > a$, and $F(x) \to \infty$ as $x \to \infty$.*

6.5 The Van der Pol Oscillator

The most important Liénard system is the **Van der Pol oscillator**:

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \qquad \mu > 0. \tag{34}$$

Here $f(x) = \mu(x^2 - 1)$ and $g(x) = x$. The damping is *negative* for $|x| < 1$ (energy is pumped in) and *positive* for $|x| > 1$ (energy is dissipated). This self-regulating mechanism produces a unique, stable limit cycle for any $\mu > 0$.

Rewriting as a system with $y = \dot{x}/\mu + x^3/3 - x$ (Liénard plane):

$$\dot{x} = \mu\big(y - F(x)\big), \qquad \dot{y} = -\frac{x}{\mu}, \qquad F(x) = \frac{x^3}{3} - x. \tag{35}$$

6.6 Relaxation Oscillations

For $\mu \gg 1$ (strong nonlinearity), the Van der Pol oscillator exhibits **relaxation oscillations**: the motion consists of slow crawls along the cubic nullcline $y = F(x) = x^3/3 - x$ punctuated by fast jumps between the two branches.

Mechanism:

1. **Fast dynamics** ($\dot{x}$ is large): the trajectory jumps nearly horizontally between the left and right branches of the S-shaped nullcline.
2. **Slow dynamics** ($\dot{y}$ is small): the trajectory crawls along the nullcline until it reaches a knee (fold), then jumps again.

The period of relaxation oscillations scales as $T \approx (3 - 2\ln 2)\mu \approx 1.614\,\mu$ for large μ—it grows linearly with the nonlinearity parameter, in contrast to the nearly constant period of weakly nonlinear oscillations.

7 Bifurcations Revisited: Two-Dimensional Systems

7.1 Saddle-Node, Transcritical, and Pitchfork in 2D

In two dimensions, all three bifurcation types from Section 2 can occur. The mechanism is the same: one eigenvalue of the Jacobian passes through zero. The other eigenvalue remains nonzero, so the dynamics in the "transverse" direction is simply attracted to (or repelled from) a manifold on which the 1D bifurcation takes place.

7.2 Hopf Bifurcation

The **Hopf bifurcation** is the birth (or death) of a limit cycle from a fixed point. It occurs when a pair of complex conjugate eigenvalues crosses the imaginary axis.

Theorem 7.1 (Hopf Bifurcation Theorem—Informal)**.** *Consider $\dot{\vec{x}} = \vec{F}(\vec{x};\mu)$ with a fixed point at the origin. Suppose the Jacobian at the origin has eigenvalues $\lambda(\mu) = \alpha(\mu) \pm i\omega(\mu)$. If:*

1. *At $\mu = 0$: $\alpha(0) = 0$ and $\omega(0) \neq 0$ (pure imaginary eigenvalues);*
2. *$\alpha'(0) \neq 0$ (eigenvalues cross the imaginary axis with nonzero speed);*

then a Hopf bifurcation occurs at $\mu = 0$.

7.2.1 Supercritical Hopf Bifurcation

A **stable limit cycle** is born from the fixed point as it loses stability.

Normal Form: Supercritical Hopf

In polar coordinates:

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega + br^2. \tag{36}$$

- For $\mu < 0$: origin is a stable spiral.
- For $\mu > 0$: origin becomes unstable; a stable limit cycle appears at $r = \sqrt{\mu}$.

The amplitude of the limit cycle grows as $\sqrt{\mu}$, and its frequency is $\omega + b\mu$ (the parameter b shifts the frequency, while ω sets the base frequency).

7.2.2 Subcritical Hopf Bifurcation

An **unstable** limit cycle shrinks down to the origin and is absorbed as the origin gains stability.

Normal form in polar coordinates:

$$\dot{r} = \mu r + r^3, \qquad \dot{\theta} = \omega + br^2. \tag{37}$$

- For $\mu < 0$: an unstable limit cycle at $r = \sqrt{-\mu}$ coexists with a stable origin.
- At $\mu = 0$: the limit cycle collapses into the origin.
- For $\mu > 0$: the origin is unstable and no small-amplitude limit cycle exists. Trajectories may jump to a distant attractor, leading to a **hard** or **catastrophic** transition.

7.3 Global Bifurcations of Cycles

Limit cycles can also be born or destroyed through *global* bifurcations that involve large-scale rearrangements of the phase portrait.

1. **Saddle-node bifurcation of cycles (fold of cycles):** A stable and an unstable limit cycle collide and annihilate.
2. **Homoclinic bifurcation:** A limit cycle grows until it touches a saddle point, forming a *homoclinic orbit* (a trajectory that connects the saddle to itself). As the parameter passes through the bifurcation, the cycle's period diverges ($T \to \infty$), and the cycle disappears.
3. **Infinite-period (SNIPER) bifurcation:** A limit cycle passes through a saddle-node on a circle. The period diverges but the amplitude remains finite.

7.4 Hysteresis in the Josephson Junction (2D)

With capacitance included, the Josephson junction becomes a second-order system:

$$i = \sin\phi + \alpha\dot{\phi} + \ddot{\phi}, \tag{38}$$

which is equivalent to the **damped driven pendulum**:

$$\dot{\phi} = y, \qquad \dot{y} = i - \sin\phi - \alpha y. \tag{39}$$

The Jacobian is $J = \left(\begin{smallmatrix} 0 & 1 \\ -\cos\phi & -\alpha \end{smallmatrix}\right)$.

The parameter α controls the damping:

- **Overdamped** ($\alpha \gg 1$): reduces to the 1D model of Section 3.4.
- **Underdamped** ($\alpha < 1$): hysteresis occurs. As i increases past 1, the zero-voltage state is lost; but when i is decreased, the system remains on the rotating (voltage) branch until i drops below a retrapping current $i_r < 1$.

7.5 Coupled Oscillators and Quasiperiodicity

Consider two oscillators with phase variables θ_1, θ_2 on the circle, coupled weakly:

$$\dot{\theta}_1 = \omega_1 + K_1 \sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = \omega_2 + K_2 \sin(\theta_1 - \theta_2). \tag{40}$$

Defining the phase difference $\phi = \theta_1 - \theta_2$:

$$\dot{\phi} = (\omega_1 - \omega_2) - (K_1 + K_2)\sin\phi. \tag{41}$$

This is again a flow on the circle!

- **Phase locking:** If $|\omega_1 - \omega_2| < K_1 + K_2$, stable fixed points exist: the oscillators lock to a constant phase difference.
- **Phase drift:** If $|\omega_1 - \omega_2| > K_1 + K_2$, no fixed points; ϕ drifts and the oscillators are **not synchronized**.

7.5.1 Dynamics on the Torus

The combined phase space (θ_1, θ_2) lives on a **torus** $\mathbb{T}^2 = S^1 \times S^1$. When uncoupled ($K_1 = K_2 = 0$), each oscillator moves independently, and the trajectory on the torus is either:

- A **closed orbit** if ω_1/ω_2 is rational, or
- A **quasiperiodic orbit** (densely filling the torus) if ω_1/ω_2 is irrational.

7.6 Poincaré Maps

Definition 7.1 (Poincaré Map)**.** Choose a **section** Σ (a codimension-1 surface transverse to the flow) in the phase space. The **Poincaré map** (or first-return map) $P : \Sigma \to \Sigma$ sends each point $\vec{x} \in \Sigma$ to the next point where the trajectory through $\vec{x}$ intersects Σ.

This reduces a continuous-time system to a discrete-time map, lowering the dimension by one. A **closed orbit** of the flow corresponds to a **fixed point** of the Poincaré map.

8 Weakly Nonlinear Oscillations

8.1 Motivation

For the Van der Pol oscillator with small $\mu \ll 1$:

$$\ddot{x} + x = \mu(1 - x^2)\dot{x},$$

the system is a *weakly perturbed* harmonic oscillator. Regular perturbation theory (expanding $x = x_0 + \mu x_1 + \cdots$) fails because secular terms grow like $t \cos t$, making the expansion invalid for large t. The method of **two-timing** (multiple scales) resolves this by introducing two time scales.

8.2 Two-Timing (Multiple Scale Analysis)

Introduce a slow time $T = \mu t$ alongside the fast time t. Write $x = x(t, T)$ and use $\dot{x} = \partial_t x + \mu \partial_T x$.

Order μ^0:

$$\partial_t^2 x_0 + x_0 = 0 \implies x_0 = r(T)\cos\big(t + \phi(T)\big).$$

The amplitude r and phase ϕ are allowed to vary on the slow time scale T.

Order μ^1: Substituting into the Van der Pol equation and collecting the μ^1 terms yields:

$$\partial_t^2 x_1 + x_1 = \left[\dot{r} - \tfrac{r}{2}(1 - \tfrac{r^2}{4})\right] \cdot 2\sin(t + \phi) + (\text{non-resonant terms}).$$

To suppress secular growth, set the coefficient of the resonant term to zero:

$$\frac{dr}{dT} = \frac{r}{2}\left(1 - \frac{r^2}{4}\right), \qquad \frac{d\phi}{dT} = 0. \tag{42}$$

Amplitude equation. Equation (42) is a logistic-type ODE: $r^* = 0$ (unstable) and $r^* = 2$ (stable). In original time, the limit cycle has amplitude $r = 2$ and period $T = 2\pi$—consistent with the known exact result for the Van der Pol oscillator.

8.3 Averaging Method

A complementary approach is **averaging**. Write the slowly varying amplitude in polar form $x = r\cos\theta$, $\dot{x} = -r\sin\theta$, where $\theta = t + \phi$. The exact transformation to (r, ϕ) gives:

$$\dot{r} = \mu f(r\cos\theta, -r\sin\theta) \cdot (-\sin\theta), \tag{43}$$

$$r\dot{\phi} = \mu f(r\cos\theta, -r\sin\theta) \cdot (-\cos\theta). \tag{44}$$

For small μ, r and ϕ change slowly over one cycle. **Averaging** over $\theta \in [0, 2\pi]$ gives the **averaged equations**:

$$\dot{r} = \frac{\mu}{2\pi}\int_0^{2\pi} f(r\cos\theta, -r\sin\theta)(-\sin\theta)\,d\theta, \quad \dot{\phi} = \frac{\mu}{2\pi r}\int_0^{2\pi} f(r\cos\theta, -r\sin\theta)(-\cos\theta)\,d\theta. \tag{45}$$

These averaged equations are easier to analyze than the full system and capture the slow dynamics accurately to $O(\mu)$ over times $t = O(1/\mu)$.

Example 8.1 (Averaging for Van der Pol)**.** With $f = (1 - x^2)\dot{x} = (1 - r^2\cos^2\theta)(r\sin\theta)$:

$$\dot{r} = \frac{\mu}{2\pi}\int_0^{2\pi}(1 - r^2\cos^2\theta)(r\sin\theta)(-\sin\theta)\,d\theta = \frac{\mu r}{2}\left(1 - \frac{r^2}{4}\right),$$

recovering (42). The frequency correction $\dot{\phi}$ vanishes to this order (no frequency shift at leading order for Van der Pol).

8.4 Frequency Shift (Duffing Oscillator)

Consider the weakly nonlinear Duffing oscillator:

$$\ddot{x} + x + \mu x^3 = 0.$$

By two-timing, the amplitude is constant ($r = r_0$) and the phase shifts:

$$\dot{\phi} = \frac{3\mu r_0^2}{8},$$

giving a corrected frequency $\omega = 1 + \frac{3\mu r_0^2}{8}$ (amplitude-dependent frequency, a hallmark of nonlinearity). This is the **backbone curve** of the Duffing oscillator.

Part III
Chaos

9 The Lorenz Equations

9.1 Origin and Physical Motivation

Edward Lorenz (1963) derived a simplified model of atmospheric convection (Rayleigh–Bénard convection). The system consists of three coupled ODEs:

$$\boxed{\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - y - xz, \qquad \dot{z} = xy - bz,} \tag{46}$$

where $\sigma > 0$ is the Prandtl number, $r > 0$ is (proportional to) the Rayleigh number, and $b > 0$ is a geometric factor. All parameters are positive. The classic parameter values are $\sigma = 10$, $b = 8/3$.

9.2 Physical Derivation: The Lorenz Waterwheel

A concrete physical realization of the Lorenz equations is the **chaotic waterwheel** (Malkus, 1972). Water drips at a constant rate into buckets arranged around a wheel. Each bucket has a small hole so water leaks out. The wheel can rotate freely in either direction, subject to friction.

Let θ be the angular position of a bucket. Expand the water mass distribution in Fourier modes:

$$m(\theta, t) = \sum_{n=0}^{\infty} [a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta)].$$

The forcing (inflow) is $Q(\theta) = Q_0(1-\cos\theta)$ (most water near the top). The equations for ω (angular velocity) and the $n = 1$ Fourier coefficients (a_1, b_1) close onto themselves:

$$\dot{\omega} = -\nu\omega + \frac{\pi g r}{I} a_1, \tag{47}$$

$$\dot{a}_1 = \omega b_1 - K a_1, \tag{48}$$

$$\dot{b}_1 = -\omega a_1 - K b_1 + q_1, \tag{49}$$

where ν is rotational damping, K is the leakage rate, and q_1 is the $n = 1$ inflow coefficient.

Under the substitution $x = C_1 a_1$, $y = C_2\omega$, $z = C_3(b_1 - q_1/K)$ with appropriate constants, these equations *are exactly the Lorenz equations.* This waterwheel analogy gives physical meaning to the variables: x and y are proportional to the mass imbalance and angular velocity; z measures the deviation of the mass distribution from the symmetric state.

9.3 Simple Properties

1. **Symmetry:** The system is invariant under $(x, y, z) \to (-x, -y, z)$.

2. **Dissipation:** The divergence of the flow is

$$\nabla \cdot \vec{F} = -\sigma - 1 - b < 0,$$

   so volumes in phase space contract exponentially. This means the system is **dissipative** and all trajectories eventually become confined to an attractor of zero volume.

3. **Fixed points:**

   - The origin $O = (0, 0, 0)$ exists for all r.
   - For $r > 1$, two additional fixed points appear via a **supercritical pitchfork** bifurcation:

$$C^{\pm} = \big(\pm\sqrt{b(r-1)},\ \pm\sqrt{b(r-1)},\ r-1\big).$$

4. **Stability of the origin:** The eigenvalues at O are found from the Jacobian. The origin is stable for $r < 1$ and unstable (a saddle with one unstable direction) for $r > 1$.

5. **Stability of $C^{\pm}$:** The Jacobian at C^{+} is

$$J = \begin{pmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & -x^* \\ x^* & x^* & -b \end{pmatrix}\Bigg|_{C^+}, \quad x^* = \sqrt{b(r-1)}.$$

The characteristic polynomial is cubic. By the Routh–Hurwitz criterion, $C^{\pm}$ are stable for $1 < r < r_H$, where

$$r_H = \sigma \frac{\sigma + b + 3}{\sigma - b - 1} \approx 24.74 \quad (\sigma = 10,\, b = 8/3). \tag{50}$$

(This requires $\sigma > b + 1$, which holds for the classic values.) At $r = r_H$, a **subcritical Hopf bifurcation** occurs: $C^{\pm}$ lose stability and an unstable limit cycle is absorbed.

6. **Trapping region:** Using the Liapunov-like function $V = rx^2 + \sigma y^2 + \sigma(z - 2r)^2$, one can show $\dot{V} < 0$ outside a large ellipsoid. Hence all trajectories eventually enter this region and remain bounded.

9.4 Chaos on the Strange Attractor

For $r > r_H$ (and the classic parameters), trajectories exhibit **chaotic** behavior:

- **Aperiodic:** Trajectories never repeat, even though they are bounded.
- **Sensitive dependence on initial conditions:** Nearby trajectories diverge exponentially. This is the hallmark of chaos.
- **Strange attractor:** The attractor has a fractal structure with zero volume but infinite detail.

Definition 9.1 (Sensitive Dependence and Horizon Time)**.** Two nearby initial conditions separated by δ_0 diverge roughly as $\delta(t) \sim \delta_0\, e^{\lambda t}$, where $\lambda > 0$ is the **Lyapunov exponent**. Predictions become unreliable when $\delta(t) \sim \epsilon$ (some tolerance). The **horizon time** is:

$$t_h \sim \frac{1}{\lambda} \ln \frac{\epsilon}{\delta_0}. \tag{51}$$

Beyond t_h, prediction is essentially impossible despite deterministic dynamics. Note the logarithmic dependence: even dramatic improvements in measurement precision δ_0 yield only modest increases in t_h.

9.5 The Lorenz Map

Lorenz discovered that the chaotic dynamics can be partially understood by a one-dimensional map. Record the successive local maxima z_n of $z(t)$. Plotting z_{n+1} vs. z_n yields a remarkably thin, single-humped curve (approximately a tent map).

Key observations:

- The slope $|f'| > 1$ everywhere on the Lorenz map, implying that nearby orbits are stretched apart at each return.
- There are no stable fixed points or periodic orbits of the map, explaining the aperiodic behavior.
- The thinness of the map (nearly 1D) arises from the extreme contraction in the dissipative system: volumes contract at rate $e^{-(\sigma+1+b)t}$, so trajectories are squeezed onto an approximately 2D surface.

Why does the Lorenz map prove chaos? If the 1D map $z_{n+1} = g(z_n)$ has $|g'| > 1$ everywhere (slopes steeper than 1 throughout), then nearby initial conditions separate exponentially at each return. A fixed-point z^* of g would be unstable ($|g'(z^*)| > 1$), confirming no stable

periodic orbits. The sensitive dependence is then rigorously demonstrated by the positive Lyapunov exponent of the map:

$$\lambda_{\text{map}} = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |g'(z_k)| > 0.$$

9.6 Exploring Parameter Space of the Lorenz System

As r increases (with $\sigma = 10$, $b = 8/3$), the Lorenz system passes through several regimes:

1. $0 < r < 1$: The origin is the global attractor. All motion decays.
2. $1 < r < r_H \approx 24.74$: The fixed points $C^{\pm}$ are stable spirals. Transient chaos may occur for $r > 13.9$ (where a "strange saddle" exists), but all trajectories eventually settle to C^+ or C^-.
3. $r_H < r$: Chaotic attractor (the "butterfly"). The system exhibits sustained aperiodic motion, sensitive dependence, and a fractal attractor.
4. Periodic windows exist within the chaotic regime, where stable periodic orbits briefly emerge.

9.7 The AC-Driven Pendulum and Routes to Chaos

The damped pendulum with a periodic (AC) driving torque provides another canonical route to chaos:

$$\Gamma \cos(\omega t) = \ddot{\theta} + b\dot{\theta} + \sin\theta. \tag{52}$$

Writing as a system with $\dot{\theta} = v$, $\dot{v} = -bv - \sin\theta + \Gamma\cos(\omega y)$, $\dot{y} = 1$, this is a three-dimensional autonomous system (the explicit time dependence elevates it to 3D).

The Poincaré section (strobing at the drive frequency) reduces it to a 2D map. As Γ increases, the system passes through:

- Periodic (phase-locked) motion,
- Period-doubling cascades,
- Chaotic tumbling,

providing a concrete physical system exhibiting the period-doubling route to chaos.

10 Lyapunov Exponents

10.1 Definition for Continuous-Time Systems

For a flow $\dot{\vec{x}} = \vec{F}(\vec{x})$ in $\mathbb{R}^n$, the **Lyapunov exponents** quantify the average rate of exponential divergence (or convergence) of infinitesimally close trajectories.

Let $\vec{x}(t)$ be a reference trajectory and $\vec{x}(t) + \delta\vec{x}(t)$ a nearby one. The perturbation $\delta\vec{x}$ satisfies the **variational equations**:

$$\frac{d}{dt}\delta\vec{x} = J\big(\vec{x}(t)\big)\,\delta\vec{x}, \tag{53}$$

where $J(\vec{x}) = D\vec{F}(\vec{x})$ is the Jacobian matrix evaluated along the reference orbit. Note that J is *time-dependent* because $\vec{x}(t)$ changes.

Definition 10.1 (Lyapunov Exponent—Continuous Time)**.** The i**-th Lyapunov exponent** is

$$\lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\delta\vec{x}_i(t)\|}{\|\delta\vec{x}_i(0)\|}, \tag{54}$$

where $\delta\vec{x}_i(0)$ is chosen appropriately (along the i-th most unstable direction). By convention, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

Computing multiple exponents requires orthogonalizing the perturbation vectors at each step (Gram–Schmidt), since without this all vectors tend to align with the most unstable direction.

10.2 The Lyapunov Spectrum

For an n-dimensional flow, there are n Lyapunov exponents. Their properties constrain the dynamics:

Properties of the Lyapunov Spectrum

1. For a **dissipative** system, the sum of all Lyapunov exponents equals the average divergence of the flow:
$$\sum_{i=1}^{n} \lambda_i = \lim_{T\to\infty} \frac{1}{T} \int_0^T \nabla \cdot \vec{F}\big(\vec{x}(t)\big)\, dt. \tag{55}$$
For the Lorenz system, $\nabla\cdot\vec{F} = -(\sigma+1+b) < 0$, so $\lambda_1+\lambda_2+\lambda_3 = -(\sigma+1+b) = -13.667$.
2. For a **flow** (continuous time), at least one exponent is zero: $\lambda_j = 0$ corresponds to perturbations along the flow direction, which neither grow nor shrink.
3. A **fixed point** has all $\lambda_i < 0$.
4. A **stable limit cycle** has one zero exponent (along the orbit) and all others negative.
5. A **chaotic attractor** has $\lambda_1 > 0$, $\lambda_j = 0$ (flow direction), and the remaining $\lambda_i < 0$.

10.3 Lyapunov Exponents for the Lorenz System

For the classic parameters $(\sigma, b, r) = (10,\ 8/3,\ 28)$, numerical computation gives:

$$\lambda_1 \approx +0.906, \qquad \lambda_2 = 0, \qquad \lambda_3 \approx -14.572. \tag{56}$$

Checking: $\lambda_1 + \lambda_2 + \lambda_3 \approx -13.666 \approx -(\sigma + 1 + b)$. ✓

The positive λ_1 confirms chaos. The horizon time (Section 9) uses λ_1: errors double roughly every $\ln 2/\lambda_1 \approx 0.77$ time units.

10.4 Kaplan–Yorke (Lyapunov) Dimension

The fractal dimension of a strange attractor can be estimated from the Lyapunov spectrum via the **Kaplan–Yorke conjecture**:

Kaplan–Yorke Dimension

Order the exponents $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Let j be the largest index such that $\lambda_1 + \lambda_2 + \cdots + \lambda_j \geq 0$. Then

$$d_{\mathrm{KY}} = j + \frac{\lambda_1 + \cdots + \lambda_j}{|\lambda_{j+1}|}. \tag{57}$$

For the Lorenz attractor: $d_{\mathrm{KY}} = 2 + \lambda_1/|\lambda_3| = 2 + 0.906/14.572 \approx 2.062$. This agrees well with direct box-counting measurements (≈ 2.06), supporting the conjecture.

Interpretation of d_{KY}

The integer part $j = 2$ means the attractor locally "looks like" a two-dimensional surface. The fractional part 0.062 reflects the fine layered (Cantor-set) structure in the transverse direction. The attractor is slightly thicker than a surface but far thinner than a solid.

11 What Is Chaos?

Before proceeding to maps, let us formalize the notion of chaos.

Definition 11.1 (Chaos—Working Definition)**.** A dynamical system is **chaotic** if it satisfies all three conditions:

1. **Aperiodic long-term behavior:** Trajectories do not settle to fixed points, periodic orbits, or quasiperiodic orbits.
2. **Deterministic:** There is no randomness in the equations—the system is governed by definite rules with no stochastic terms.
3. **Sensitive dependence on initial conditions:** Nearby trajectories diverge exponentially, with at least one positive Lyapunov exponent.

Remark 11.1. Chaos is *not* randomness. The underlying equations are perfectly deterministic; the apparent randomness arises from the amplification of microscopic uncertainties in initial conditions. This is sometimes called **deterministic chaos** or **deterministic aperiodicity**.

Remark 11.2. Chaos requires at least three dimensions for continuous-time (flow) systems. In two dimensions, the Poincaré–Bendixson theorem constrains bounded trajectories to approach fixed points or closed orbits. However, for discrete-time systems (maps), chaos is possible even in one dimension (e.g., the logistic map).

12 Routes to Chaos

Three principal routes by which an ordered system transitions to chaos have been identified theoretically and observed experimentally.

12.1 Route 1: Period-Doubling Cascade (Feigenbaum)

The most common route begins with a stable fixed point (or limit cycle), which undergoes a **flip bifurcation** ($f'(x^*) = -1$ for maps, or a subcritical Hopf in flows) as a parameter r is increased.

A period-2 orbit appears; it in turn undergoes a flip, producing period-4; then period-8; and so on in an infinite cascade.

Structure of the cascade. Let r_n be the parameter value at which the 2^n-cycle is born.

- The sequence $\{r_n\}$ converges geometrically: $r_n \to r_\infty$ as $n \to \infty$, with

$$\frac{r_n - r_{n-1}}{r_{n+1} - r_n} \longrightarrow \delta \approx 4.669\,201\,609\ldots \tag{58}$$

  (Feigenbaum's constant δ).

- The amplitudes d_n of the 2^n-cycle near r_∞ also scale:

$$\frac{d_n}{d_{n+1}} \longrightarrow \alpha \approx 2.502\,907\ldots \tag{59}$$

  (Feigenbaum's α).

- Both δ and α are **universal**: they are the same for every map with a single smooth (quadratic) maximum, regardless of the specific functional form. This was Feigenbaum's 1978 discovery.

Why universality? The renormalization-group explanation (Section 13.7) identifies δ and α as eigenvalues of the doubling operator at its fixed point g. The fixed point is a universal object independent of the particular map.

Onset of chaos. At $r = r_\infty$, the Lyapunov exponent $\lambda = 0$. For r slightly above r_∞:

$$\lambda \sim (r - r_\infty)^\nu, \qquad \nu \approx 0.4498\ldots$$

providing another universal scaling law.

Experimental realizations.

- Rayleigh–Bénard convection (Libchaber & Maurer, 1980): observed period-doubling cascade in liquid helium.
- Driven nonlinear circuits (Linsay, 1981).
- Dripping faucets, chemical oscillators (Belousov–Zhabotinsky reaction).

12.2 Route 2: Quasiperiodic Route (Ruelle–Takens–Newhouse)

As a parameter varies, the system passes through:

$$\text{fixed point} \;\to\; \text{limit cycle} \;\to\; \text{2-torus} \;\to\; \text{chaos}.$$

1. A **Hopf bifurcation** creates a limit cycle with frequency ω_1.
2. A second Hopf bifurcation (Neimark–Sacker for maps) creates a **torus** $\mathbb{T}^2$ with a second frequency ω_2. If ω_1/ω_2 is irrational, motion is quasiperiodic (a trajectory densely fills the torus).
3. A third Hopf (Landau–Hopf scenario) would add ω_3 on a 3-torus. However, Ruelle, Takens, and Newhouse (1978) proved that a 3-torus is **not structurally stable**: any small perturbation can destroy it and produce a **strange attractor**.

Thus, two incommensurate frequencies give quasiperiodicity; but as soon as a third mode appears, chaos can emerge without further parameter changes.

Power Spectrum Signature

In the quasiperiodic route, the power spectrum shows two sharp peaks at ω_1 and ω_2 (and their combinations $m\omega_1 + n\omega_2$). The onset of chaos is marked by the appearance of broad-band noise in the spectrum. This can be distinguished from the period-doubling route (which shows peaks at subharmonics $\omega/2$, $\omega/4$, ...).

12.3 Route 3: Intermittency (Pomeau–Manneville, 1980)

In intermittency, a **periodic orbit disappears** at a bifurcation but leaves a "ghost" through which trajectories pass nearly periodically for long stretches before being ejected into chaotic bursts.

Three types, based on the local bifurcation:

1. **Type I** (saddle-node of cycles): A stable and unstable periodic orbit collide and annihilate. For r slightly past the bifurcation, the orbit acts as a bottleneck. Mean laminar length:
$$\langle \tau \rangle \sim (r - r_c)^{-1/2}.$$
2. **Type II** (subcritical Hopf): Complex conjugate Floquet multipliers cross the unit circle. Laminar phases are longer and more structured.
3. **Type III** (subcritical period-doubling): A Floquet multiplier exits at -1. Mean laminar length:
$$\langle \tau \rangle \sim (r_c - r)^{-1}.$$

Phenomenology of Type I. Consider a 1D map with a near tangency: $x_{n+1} = x_n + \epsilon + x_n^2$ for small $\epsilon > 0$. The trajectory passes slowly through the narrow gap near $x = 0$ ("laminar phase"), taking $\sim \pi/\sqrt{\epsilon}$ iterations, then escapes to a chaotic region ("turbulent burst"), re-injects, and repeats.

Experimental realization. Pomeau and Manneville identified intermittency in a Lorenz-type model. It has also been observed in Rayleigh–Bénard convection and in the logistic map near the periodic windows (e.g., the period-3 window at $r \approx 3.83$).

12.4 Summary: Signatures of the Three Routes

Route	Bifurcation	Power spectrum	Scaling
Period-doubling	Flip cascade	Subharmonics $\omega/2^n$	$\delta \approx 4.67$
Quasiperiodic	Hopf×2+RTN	Broad-band after torus	—
Intermittency	Tangent/Hopf/PD	Broad-band throughout	$\langle \tau \rangle \sim \epsilon^{-1/2}$

13 One-Dimensional Maps

13.1 Introduction

Motivated by Poincaré maps and the Lorenz map, we now study discrete-time dynamical systems (**iterated maps**) of the form
$$x_{n+1} = f(x_n), \tag{60}$$
where $f : \mathbb{R} \to \mathbb{R}$ is a given function.

13.2 Fixed Points and Cobweb Diagrams

Definition 13.1. A point x^* is a **fixed point** of the map (60) if $f(x^*) = x^*$.

Graphically, fixed points are found by intersecting the graph of $f(x)$ with the diagonal line $y = x$.

Stability of a Fixed Point (Maps)

Let x^* be a fixed point of $x_{n+1} = f(x_n)$. The **multiplier** is $\mu = f'(x^*)$.

- $|\mu| < 1$: x^* is **stable** (attracting).
- $|\mu| > 1$: x^* is **unstable** (repelling).
- $|\mu| = 1$: inconclusive.

Proof. Let $\delta_n = x_n - x^*$. Then $\delta_{n+1} = f(x^* + \delta_n) - x^* \approx f'(x^*)\,\delta_n$, so $\delta_n \approx [f'(x^*)]^n \delta_0$. □

Cobweb construction: A graphical method for iterating maps. Starting from $(x_0, 0)$:

1. Go vertically to $y = f(x_0)$.
2. Go horizontally to the diagonal $y = x$ (so the horizontal coordinate is now $x_1 = f(x_0)$).
3. Repeat.

13.3 Periodic Orbits

Definition 13.2. A point p has **period** n if $f^n(p) = p$ and $f^k(p) \neq p$ for $0 < k < n$, where f^n denotes the n-fold composition $f \circ f \circ \cdots \circ f$.

A period-n orbit is a fixed point of f^n. Its stability is determined by the multiplier of f^n:

$$(f^n)'(p) = f'(p) \cdot f'(f(p)) \cdot f'(f^2(p)) \cdots f'(f^{n-1}(p)) = \prod_{k=0}^{n-1} f'(f^k(p)). \tag{61}$$

The orbit is stable if $|(f^n)'(p)| < 1$ and unstable if $|(f^n)'(p)| > 1$.

13.4 The Logistic Map

The most studied one-dimensional map is the **logistic map**:

$$x_{n+1} = r x_n (1 - x_n), \qquad 0 \leq r \leq 4. \tag{62}$$

13.4.1 Fixed Points

Setting $x = rx(1 - x)$: the fixed points are $x^* = 0$ and $x^* = (r - 1)/r$ (for $r > 1$).

Stability of $x^* = (r - 1)/r$: $f'(x) = r - 2rx$, so $f'(x^*) = r - 2(r - 1) = 2 - r$. This fixed point is stable when $|2 - r| < 1$, i.e., $1 < r < 3$.

13.4.2 Period Doubling

At $r = 3$, the fixed point $x^* = (r-1)/r$ loses stability through a **flip bifurcation** ($f'(x^*) = -1$). A stable period-2 cycle is born. As r increases further:

- $r \approx 3$: period-2 cycle born.
- $r \approx 3.449$: period-4 cycle born.
- $r \approx 3.544$: period-8 cycle born.
- The cascade continues: period 2^n orbits for $n = 1, 2, 3, \ldots$
- The intervals between successive doublings shrink geometrically.
- At $r_\infty \approx 3.5699$: all periods 2^n coexist—the onset of chaos.

13.4.3 Period-Doubling Cascade and Feigenbaum's Universality

Let r_n be the parameter value at which the period-2^n cycle is born. Feigenbaum (1978) discovered that the ratios

$$\frac{r_n - r_{n-1}}{r_{n+1} - r_n} \to \delta \approx 4.669\,201\ldots \tag{63}$$

converge to a universal constant δ (**Feigenbaum's constant**), independent of the specific map f, as long as f has a single quadratic maximum.

13.4.4 Periodic Windows

For $r > r_\infty$, chaos is interspersed with **periodic windows**— intervals of r where a stable periodic orbit exists. The most prominent is the period-3 window near $r \approx 3.83$.

Theorem 13.1 (Li–Yorke Theorem (1975))**.** *If a continuous map on an interval has a period-3 orbit, then it has periodic orbits of* every *period.*

Theorem 13.2 (Sharkovsky's Theorem (1964))**.** *Order the positive integers by the* ***Sharkovsky ordering****:*

$$3 \rhd 5 \rhd 7 \rhd \cdots \rhd 2\cdot 3 \rhd 2\cdot 5 \rhd \cdots \rhd 4\cdot 3 \rhd 4\cdot 5 \rhd \cdots \rhd 2^3 \rhd 2^2 \rhd 2 \rhd 1.$$

(Odd numbers > 1 first, then $2\times$odds, then $4\times$odds, ..., then powers of 2 in decreasing order, ending with 1.)

If a continuous map $f : \mathbb{R} \to \mathbb{R}$ has a period-n orbit, then it also has a period-m orbit for every m with $n \rhd m$.

Remark 13.1. Period 3 is the largest in the Sharkovsky ordering, so it implies all periods. Period 2^k (for any k) implies only periods that are powers of 2 up to 2^k—this is why period-doubling cascades come in powers of 2 and terminate in "period-3 implies all" chaos.

13.4.5 Invariant Measure of the Logistic Map at $r = 4$

The logistic map $x_{n+1} = 4x_n(1 - x_n)$ at $r = 4$ preserves the **arc-sine distribution**:

$$\rho(x) = \frac{1}{\pi\sqrt{x(1-x)}}, \qquad x \in (0, 1). \tag{64}$$

This can be verified using the conjugacy $h(x) = \sin^2(\pi x/2)$ with the tent map (which preserves the uniform measure): pushing forward the uniform measure on $[0, 1]$ via h gives the arc-sine measure.

Consequence: For typical initial conditions x_0 (uniformly random), time-averages of observables $A(x_n)$ converge to spatial averages:

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} A(x_n) = \int_0^1 A(x)\,\rho(x)\,dx.$$

This is **ergodicity**.

13.5 Lyapunov Exponent

Definition 13.3 (Lyapunov Exponent for Maps)**.** For a map $x_{n+1} = f(x_n)$ with initial condition x_0, the **Lyapunov exponent** is

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln|f'(x_k)| = \lim_{n\to\infty} \frac{1}{n} \ln\left|\frac{df^n}{dx}\bigg|_{x_0}\right|. \tag{65}$$

Interpretation:

- $\lambda > 0$: nearby orbits diverge exponentially—the system is **chaotic**.
- $\lambda < 0$: nearby orbits converge—a stable periodic orbit attracts.
- $\lambda = 0$: marginal case (e.g., at a bifurcation point or neutral stability).

For two nearby initial conditions x_0 and $\tilde{x}_0 = x_0 + \delta_0$:

$$|\delta_n| \approx |\delta_0|\,e^{\lambda n} \quad \Longrightarrow \quad \lambda \approx \frac{1}{n} \ln \frac{|\delta_n|}{|\delta_0|}.$$

13.6 The Schwarzian Derivative

Definition 13.4 (Schwarzian Derivative)**.** For a C^3 function f, the **Schwarzian derivative** is

$$Sf(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2}\left(\frac{f''(x)}{f'(x)}\right)^2. \tag{66}$$

Theorem 13.3 (Singer's Theorem, 1978)**.** *If $Sf(x) < 0$ for all x in the interval, then:*

1. *Every stable periodic orbit of f attracts at least one critical point.*
2. *The number of stable periodic orbits is at most the number of critical points of f.*

For the logistic map $f(x) = rx(1 - x)$: $Sf(x) = -6r^2(1 - 2x)^2/[r(1 - 2x)]^2 - \cdots = -\frac{3}{2} < 0$ everywhere (after computation). Hence the logistic map has at most one stable periodic orbit at any parameter value—the orbit observed numerically (the period-2^n cycle in the doubling cascade, or the period-3 cycle in its window) is the *only* stable orbit.

13.7 Renormalization

The universality of the period-doubling cascade can be understood through **renormalization group** ideas.

The key insight: near the period-doubling accumulation point r_∞, the map f^{2^n} (restricted to a small interval near the critical point) looks approximately like a rescaled copy of f itself. This self-similarity leads to a functional equation (the **Feigenbaum equation**):

$$g(x) = -\alpha\, g\big(g(-x/\alpha)\big), \tag{67}$$

where g is the universal limiting function and $\alpha \approx 2.5029$ is the rescaling factor. The constant δ (Feigenbaum's constant) arises as the relevant eigenvalue of the renormalization operator linearized about g.

14 Symbolic Dynamics and the Tent Map

14.1 The Tent Map

The **tent map** is

$$T(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2}, \\ 2(1-x), & \frac{1}{2} < x \le 1. \end{cases} \tag{68}$$

The slope is ± 2 everywhere, so $|T'| = 2 > 1$ everywhere (except $x = \frac{1}{2}$).

Exact Lyapunov exponent. Since $|T'(x)| = 2$ for all x (a.e.):

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |T'(x_k)| = \ln 2 \approx 0.693.$$

The tent map is maximally chaotic in the sense that every bit of information is generated per iterate.

Invariant measure. The tent map preserves the uniform (Lebesgue) measure on $[0,1]$: if x_0 is chosen uniformly, then every x_n is also uniformly distributed.

14.2 Symbolic Dynamics

Symbolic dynamics converts the dynamics of a map into a discrete sequence problem.

Definition 14.1 (Itinerary)**.** Partition $[0,1]$ into two **symbols**: $L = [0, \frac{1}{2})$ and $R = [\frac{1}{2}, 1]$. The **itinerary** of x_0 is the bi-infinite (or one-sided) sequence

$$s(x_0) = s_0 s_1 s_2 \cdots, \quad s_n = \begin{cases} L & T^n(x_0) < \frac{1}{2}, \\ R & T^n(x_0) \ge \frac{1}{2}. \end{cases}$$

Definition 14.2 (Shift Map)**.** The **shift map** σ acts on sequences:

$$\sigma(s_0 s_1 s_2 \cdots) = s_1 s_2 s_3 \cdots.$$

The key relationship is:

$$s(T(x)) = \sigma(s(x)). \tag{69}$$

Thus the tent map dynamics on $[0,1]$ is *conjugate* to the shift map on the space of binary sequences $\{L, R\}^{\mathbb{N}}$.

Consequences of symbolic dynamics.

1. **Periodic orbits:** Period-n orbits of T correspond to period-n sequences of the shift. There are exactly 2^n sequences of period n, hence the tent map has 2^n period-n points. This confirms the existence of periodic orbits of all periods.

2. **Sensitive dependence:** Two orbits with itineraries differing first at position n are initially within 2^{-n} of each other but have diverged to $O(1)$ separation by step n. This is sensitive dependence made rigorous.

3. **Topological entropy:** The rate of growth of distinct itineraries is $h = \ln 2$, confirming maximum complexity.

14.3 Topological Conjugacy

Definition 14.3 (Topological Conjugacy)**.** Two maps $f : X \to X$ and $g : Y \to Y$ are **topologically conjugate** if there exists a homeomorphism $h : X \to Y$ such that

$$h \circ f = g \circ h, \quad \text{i.e.,} \quad h(f(x)) = g(h(x)). \tag{70}$$

Conjugate maps have the same dynamics up to relabeling of points: same number of periodic orbits of each period, same Lyapunov exponent, same topological entropy.

Theorem 14.1 (Logistic–Tent Conjugacy)**.** *The logistic map $f(x) = 4x(1-x)$ on $[0,1]$ is topologically conjugate to the tent map T on $[0,1]$ via the homeomorphism*

$$h(x) = \sin^2\!\left(\tfrac{\pi}{2}x\right), \quad h^{-1}(y) = \tfrac{2}{\pi}\arcsin\sqrt{y}. \tag{71}$$

Proof. One verifies $f(h(x)) = 4\sin^2\!\left(\frac{\pi x}{2}\right)\cos^2\!\left(\frac{\pi x}{2}\right) = \sin^2(\pi x) = h(T(x))$ using the double-angle formula $\sin(\pi x) = 2\sin\!\left(\frac{\pi x}{2}\right)\cos\!\left(\frac{\pi x}{2}\right)$. □

Corollary 14.2. *The logistic map at $r = 4$ has Lyapunov exponent $\lambda = \ln 2$ and is chaotic on all of $[0,1]$.*

14.4 Smale Horseshoe

The **Smale horseshoe** (1967) is the prototypical mechanism producing chaos in two-dimensional maps. Consider the unit square $Q = [0,1]^2$. The horseshoe map F acts as follows:

1. **Stretch** Q vertically by factor > 2 and contract horizontally by factor $< \frac{1}{2}$.

2. **Fold** the resulting rectangle into a horseshoe shape and place it back over Q.

The intersection $Q \cap F(Q)$ consists of two vertical strips. $Q \cap F(Q) \cap F^2(Q)$ consists of four strips; and so on.

Invariant set. Points that remain in Q under all forward and backward iterations form a Cantor-set-like invariant set $\Lambda = \bigcap_{n=-\infty}^{\infty} F^n(Q)$, which is homeomorphic to $\{0,1\}^{\mathbb{Z}}$ (bi-infinite binary sequences). The dynamics on Λ is conjugate to the **full two-sided shift**, which has:

- Periodic orbits of every period,
- A dense orbit (transitive dynamics),
- Sensitive dependence on initial conditions.

Horseshoe in Lorenz and Poincaré Maps

The Smale horseshoe appears as the "skeleton" underlying many chaotic systems. In the Poincaré section of the Lorenz system, successive returns trace out a shape that, after topological deformation, is conjugate to a symbolic system. This explains why the Lorenz attractor has periodic orbits of all periods interleaved with chaotic motion.

15 Fractals and Strange Attractors

15.1 Fractal Dimension

Chaotic attractors typically have a **fractal** (non-integer) dimension. This reflects their infinitely detailed, self-similar structure.

Definition 15.1 (Box-Counting Dimension)**.** Cover the set S with $N(\epsilon)$ boxes of side length ϵ. The **box-counting dimension** (or Minkowski dimension) is

$$d = \lim_{\epsilon \to 0} \frac{\ln N(\epsilon)}{\ln(1/\epsilon)}. \tag{72}$$

Example 15.1 (Self-Similar Fractals)**.**

- **Cantor set:** Formed by repeatedly removing the middle third of intervals. Dimension $d = \ln 2/\ln 3 \approx 0.631$.
- **Sierpinski triangle:** Dimension $d = \ln 3/\ln 2 \approx 1.585$.
- **Koch snowflake:** Dimension $d = \ln 4/\ln 3 \approx 1.262$.

For a self-similar fractal made of N copies of itself scaled by factor r, the dimension is $d = \ln N/\ln(1/r)$.

15.2 Dimension of Self-Similar Fractals

Definition 15.2 (Self-Similarity Dimension)**.** If a set S is the union of N non-overlapping copies of itself, each scaled by a factor r (with $0 < r < 1$), then the **self-similarity dimension** is

$$d = \frac{\ln N}{\ln(1/r)}. \tag{73}$$

This coincides with the box-counting dimension for exact self-similar sets. For strange attractors (which are typically self-similar only in a statistical sense), one uses the box-counting or correlation dimension.

16 The Rössler System

16.1 Equations and Motivation

The **Rössler system** (1976) was designed to be the simplest possible chaotic flow. It has only one nonlinear term:

$$\boxed{\dot{x} = -y - z, \qquad \dot{y} = x + ay, \qquad \dot{z} = b + z(x - c),} \tag{74}$$

with classic parameters $a = b = 0.2$ and $c = 5.7$.

16.2 Fixed Points

Setting $\dot{x} = \dot{y} = \dot{z} = 0$:

$$z = -y, \quad x = -ay = ac/2 \pm \tfrac{a}{2}\sqrt{c^2 - 4b},$$
$$y_\pm = -\tfrac{c}{2a} \pm \tfrac{1}{2a}\sqrt{c^2 - 4b}.$$

For $a = b = 0.2$, $c = 5.7$: two fixed points exist approximately at $P^+ \approx (0.007, -0.035, 0.035)$ and $P^- \approx (5.693, -28.47, 28.47)$.

Linearization at $P^\pm$ shows P^+ is an unstable spiral-saddle (eigenvalues include a positive real part) and P^- is a saddle. The attractor lives near P^+.

16.3 The Rössler Attractor and Period-Doubling Route

For small c, the Rössler system has a stable limit cycle. As c increases with $a = b = 0.2$:

c	Behavior
$c \approx 4.2$	Period-1 limit cycle
$c \approx 6.0$	Period-2
$c \approx 8.5$	Period-4
$c \approx 8.7$	Period-8, cascade continues
$c \approx 9.0$	Onset of chaos
$c = 5.7$ (classic)	Chaos: "band" attractor

The Rössler attractor has a characteristic **band shape**: trajectories spiral outward in the xy-plane, are then injected back near the z-axis, fold once, and return. Because there is only one fold, the Poincaré map is approximately a one-humped map—more regular than Lorenz—and the period-doubling cascade is clearly observed.

Rössler Lyapunov Exponents (classic parameters)

$\lambda_1 \approx +0.071$, $\lambda_2 = 0$, $\lambda_3 \approx -5.39$. Kaplan–Yorke dimension: $d_{\mathrm{KY}} = 2 + 0.071/5.39 \approx 2.013$. The attractor is barely fractal (close to a 2D surface).

16.4 Comparison: Rössler vs. Lorenz

	Lorenz	**Rössler**
Nonlinear terms	2 (xz, xy)	1 (xz)
Symmetry	$\mathbb{Z}_2$: $(x, y) \to (-x, -y)$	None
Attractor	Butterfly (two wings)	Band (one wing)
Route to chaos	Via r; subcritical Hopf	Period-doubling in c
Poincaré map	Nearly 1D humped map	Clearly 1D humped
d_{KY}	≈ 2.06	≈ 2.01

17 The Hénon Map

17.1 Definition and Motivation

The **Hénon map** (1976) is the simplest two-dimensional map known to have a strange attractor:

$$x_{n+1} = 1 - ax_n^2 + y_n, \qquad y_{n+1} = bx_n, \tag{75}$$

with classic parameters $a = 1.4$, $b = 0.3$. It was constructed as a simplified model of the Poincaré section of the Lorenz system.

17.2 Properties

Area contraction. The Jacobian of the map is

$$J = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}, \quad \det J = -b.$$

The area of any region contracts by factor $|b| = 0.3$ at each iteration. Since $|b| < 1$, the map is **dissipative**.

Invertibility. $\det J \neq 0$ for all (x, y), so the map is invertible:

$$x_n = y_{n+1}/b, \qquad y_n = x_{n+1} - 1 + a(y_{n+1}/b)^2.$$

Reduction to 1D. When $b = 0$: $y_{n+1} = 0$ for all $n \geq 1$, and the map reduces to $x_{n+1} = 1 - ax_n^2$, which is conjugate to the logistic map. Thus the Hénon map "thickens" the logistic map into two dimensions.

17.3 Fixed Points and Stability

Fixed points satisfy $x^* = 1 - a(x^*)^2 + bx^*$, giving

$$x^* = \frac{(b-1) \pm \sqrt{(1-b)^2 + 4a}}{2a}.$$

For $a = 1.4$, $b = 0.3$: $x_1^* \approx 0.884$ (unstable) and $x_2^* \approx -1.026$ (unstable).

Both fixed points are unstable for the classic parameters—the attractor is not a fixed point or cycle.

17.4 The Hénon Strange Attractor

For $a = 1.4$, $b = 0.3$, a typical initial condition generates an orbit that converges to the **Hénon strange attractor**:

- **Dimension:** $d \approx 1.26$ (box-counting). Kaplan–Yorke: $\lambda_1 \approx 0.42$, $\lambda_2 \approx -1.62$, giving $d_{\mathrm{KY}} = 1 + 0.42/1.62 \approx 1.26$.
- **Structure:** Zooming in reveals a Cantor-set structure in the transverse direction—layers of curves, each containing more layers, ad infinitum.
- **Topological horseshoe:** The Hénon attractor contains a Smale horseshoe as a subsystem, proving the existence of periodic orbits of all periods.

17.5 Bifurcation Diagram of the Hénon Map

Fixing $b = 0.3$ and varying a:

- $0 < a \lesssim 0.37$: stable fixed point.
- $a \approx 0.37$: first period-doubling.
- Period-doubling cascade with Feigenbaum ratio δ (same universal constant as the logistic map).
- $a \gtrsim 1.06$: onset of sustained chaos.
- $a = 1.4$: classic strange attractor.
- $a > 1.43$: most orbits escape to infinity.

17.6 Strange Attractors

Definition 17.1 (Strange Attractor)**.** An attractor is **strange** if it has a fractal structure (non-integer dimension). A **chaotic attractor** is a strange attractor on which trajectories exhibit sensitive dependence on initial conditions.

Key examples:

- The **Lorenz attractor** has dimension ≈ 2.06—slightly more than a surface, reflecting its layered structure.
- The **Hénon map** $x_{n+1} = 1 - ax_n^2 + y_n$, $y_{n+1} = bx_n$ (with $a = 1.4$, $b = 0.3$) has a strange attractor with dimension ≈ 1.26.
- The **Rössler system** $\dot{x} = -y - z$, $\dot{y} = x + ay$, $\dot{z} = b + z(x - c)$ produces a simpler "band" attractor that is easier to visualize than Lorenz.

Stretching and Folding

Strange attractors arise from the interplay of two mechanisms:

1. **Stretching:** Nearby trajectories diverge (positive Lyapunov exponent), creating sensitive dependence.
2. **Folding:** Since the attractor is bounded, the stretched-out trajectories must fold back on themselves, creating the layered fractal structure.

This stretch-and-fold mechanism is analogous to kneading dough—each iteration creates finer and finer layers.

17.7 Correlation Dimension

The box-counting dimension counts boxes uniformly. A more practical and statistically robust measure is the **correlation dimension** d_2 (Grassberger–Procaccia, 1983).

Definition 17.2 (Correlation Integral)**.** Given N points $\{\vec{x}_i\}$ sampled from the attractor, define

$$C(\epsilon) = \lim_{N\to\infty} \frac{1}{N^2} \#\{(i,j) : \|\vec{x}_i - \vec{x}_j\| < \epsilon\}. \tag{76}$$

For a fractal attractor, $C(\epsilon) \sim \epsilon^{d_2}$ as $\epsilon \to 0$, defining the correlation dimension:

$$d_2 = \lim_{\epsilon \to 0} \frac{\ln C(\epsilon)}{\ln \epsilon}. \tag{77}$$

In practice, d_2 is estimated from the slope of $\ln C(\epsilon)$ vs. $\ln \epsilon$. For the Lorenz attractor, $d_2 \approx 2.05$, consistent with $d_{\mathrm{KY}} \approx 2.06$.

17.8 Crises: Sudden Changes in Chaotic Attractors

A **crisis** is a sudden qualitative change in a chaotic attractor caused by its collision with an unstable fixed point or periodic orbit as a parameter varies.

Boundary crisis. The chaotic attractor collides with its own *basin boundary* (which is an unstable set). After the collision, the attractor is destroyed and trajectories escape to infinity (or another attractor). The attractor existed just before the crisis; after, typical orbits are transient before escaping.

Interior crisis. Two (or more) chaotic attractors suddenly merge into one larger attractor. In the logistic map, the period-3 window ends when the chaotic band merges with a larger chaotic region— this is an interior crisis at $r \approx 3.857$.

Scaling near a crisis. Near a boundary crisis at parameter μ_c, the average transient lifetime before escape scales as:

$$\langle \tau \rangle \sim |\mu - \mu_c|^{-\gamma}, \tag{78}$$

where γ is related to the eigenvalues of the unstable fixed point at the collision.

17.9 Summary: Properties of Chaotic Attractors

Attractor	λ_1	d_{KY}	**Route**
Lorenz (std. params)	≈ 0.906	≈ 2.06	Subcritical Hopf
Rössler (std.)	≈ 0.071	≈ 2.01	Period-doubling
Hénon ($a = 1.4$)	≈ 0.42	≈ 1.26	Period-doubling
Logistic ($r = 4$)	$\ln 2 \approx 0.693$	1.0	Period-doubling

Part IV
Collective Behavior

18 The Kuramoto Model

18.1 Motivation: Synchronization in Nature

Synchronization phenomena are ubiquitous: fireflies flashing in unison, cardiac pacemaker cells, audience clapping in rhythm, circadian rhythms, and power grids. The challenge is to understand how a large population of coupled oscillators can spontaneously synchronize.

18.2 The Model

Yoshiki Kuramoto (1975) proposed a beautifully simple model. Consider N oscillators with phases $\theta_1, \ldots, \theta_N$. Each oscillator has its own natural frequency ω_i, drawn from a distribution $g(\omega)$ (typically symmetric and unimodal, e.g., Gaussian or Lorentzian). The coupling is all-to-all and sinusoidal:

$$\boxed{\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \qquad i = 1, \ldots, N.} \tag{79}$$

Here $K \geq 0$ is the **coupling strength**.

18.3 The Order Parameter

To measure the degree of synchronization, Kuramoto introduced the **complex order parameter**:

$$re^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}, \tag{80}$$

where $r \in [0, 1]$ is the **coherence** (amplitude of the mean field) and ψ is the **average phase**.

- $r = 1$: all oscillators are perfectly synchronized ($\theta_j = \psi$ for all j).
- $r = 0$: the phases are uniformly distributed around the circle (complete incoherence).
- $0 < r < 1$: partial synchronization.

Mean-field form. Multiplying both sides of (80) by $e^{-i\theta_i}$ and taking the imaginary part:

$$r\sin(\psi - \theta_i) = \frac{1}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i).$$

Thus the Kuramoto model becomes:

$$\dot{\theta}_i = \omega_i + Kr\sin(\psi - \theta_i). \tag{81}$$

Each oscillator interacts only with the "mean field" (r, ψ), not with every other oscillator individually.

18.4 Self-Consistency Analysis (Infinite N)

In the $N \to \infty$ limit, we look for a steady-state solution in a rotating frame where ψ is constant (choose a frame where $\psi = 0$). Oscillators split into two groups:

1. **Locked oscillators** ($|\omega_i| \leq Kr$): these are entrained to the mean field and satisfy $\dot{\theta}_i = 0$, i.e., $\omega_i = -Kr\sin\theta_i$, giving $\theta_i = \arcsin(\omega_i/Kr)$.
2. **Drifting oscillators** ($|\omega_i| > Kr$): these are not captured by the mean field and continue to rotate, but nonuniformly. Their steady-state density is inversely proportional to their angular velocity: $\rho(\omega, \theta) \propto 1/|\dot{\theta}|$.

The self-consistency condition requires the locked and drifting oscillators to produce exactly the assumed order parameter r. This yields:

$$r = \int_{-\pi/2}^{\pi/2} Kr\cos^2\theta\; g(Kr\sin\theta)\, d\theta, \tag{82}$$

or equivalently, after canceling r (for $r \neq 0$):

$$1 = K\int_{-\pi/2}^{\pi/2} \cos^2\theta\; g(Kr\sin\theta)\, d\theta. \tag{83}$$

18.5 Critical Coupling and the Phase Transition

Theorem 18.1 (Onset of Synchronization)**.** *There exists a critical coupling strength*

$$K_c = \frac{2}{\pi\, g(0)}, \tag{84}$$

such that:

- *For $K < K_c$: the only stable solution is $r = 0$ (complete incoherence).*
- *For $K > K_c$: a partially synchronized state with $r > 0$ emerges.*

Near the transition, $r \sim \sqrt{K - K_c}$, analogous to a ***supercritical pitchfork bifurcation*** *(or a second-order phase transition in statistical mechanics).*

Derivation of K_c. From (83), in the limit $r \to 0^+$:

$$1 = K\int_{-\pi/2}^{\pi/2} \cos^2\theta\; g(0)\, d\theta = Kg(0)\cdot\frac{\pi}{2},$$

since $g(Kr\sin\theta) \to g(0)$ uniformly and $\int_{-\pi/2}^{\pi/2}\cos^2\theta\, d\theta = \pi/2$. Solving: $K_c = 2/(\pi g(0))$. □

Analogy with Phase Transitions

The Kuramoto model undergoes a transition remarkably similar to a **ferromagnetic phase transition**: below K_c ("high temperature"), there is no order; above K_c ("low temperature"), spontaneous order emerges. The order parameter r plays the role of the magnetization, and K plays the role of inverse temperature. The square-root scaling $r \sim \sqrt{K - K_c}$ gives a critical exponent of $\beta = 1/2$, the same as mean-field theory in statistical mechanics.

Part V
Additional Topics

19 Hamiltonian Systems and KAM Theory

19.1 Hamiltonian Systems

A **Hamiltonian system** with n degrees of freedom has a function $H(q_1, \ldots, q_n, p_1, \ldots, p_n)$ (the Hamiltonian) and equations of motion:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{85}$$

Key differences from dissipative systems:

1. **Volume preservation (Liouville's theorem):** $\nabla \cdot \vec{F} = \sum_i (\partial^2 H/\partial q_i \partial p_i - \partial^2 H/\partial p_i \partial q_i) = 0$. Phase-space volumes are preserved.
2. **No attractors:** Since volume is preserved, there can be no attracting sets (which would require volume contraction).
3. **Energy conservation:** $\dot{H} = 0$ along trajectories.
4. **Time-reversibility:** Under $t \to -t$, $p \to -p$, the equations are unchanged.

19.2 Integrable Systems and Action-Angle Variables

A Hamiltonian system with n degrees of freedom is **integrable** (in the Liouville–Arnold sense) if it has n independent conserved quantities $I_1, \ldots, I_n$ in involution ($\{I_i, I_j\} = 0$ for all i, j, where $\{\cdot, \cdot\}$ is the Poisson bracket).

In that case, the motion is confined to n-dimensional tori in the $2n$-dimensional phase space. In **action-angle variables** (I_i, θ_i), the equations become:

$$\dot{I}_i = 0, \quad \dot{\theta}_i = \omega_i(I_1, \ldots, I_n),$$

so the actions are constants and the angles evolve uniformly—pure quasiperiodic motion.

19.3 Chaos in Hamiltonian Systems

For *non-integrable* Hamiltonian systems, the KAM theorem and its breakdown govern the onset of chaos.

Theorem 19.1 (KAM Theorem—Informal)**.** *Consider an integrable Hamiltonian H_0 perturbed by ϵH_1. Most tori with **sufficiently irrational** frequency ratios ω_i/ω_j survive the perturbation (with slight deformation) for small ϵ. Tori with **rational** (resonant) frequency ratios are destroyed and replaced by alternating stable/unstable periodic orbits (Poincaré–Birkhoff theorem).*

Consequences:

- The phase portrait of a non-integrable Hamiltonian is a mixture of regular (KAM tori) and chaotic (destroyed tori) regions.
- Chaos lives in the gaps between KAM tori, near the separatrices of resonant orbits.
- As ϵ increases, more and more tori break, and the chaotic region grows. At a critical value, the last KAM torus breaks and global diffusion becomes possible (Chirikov resonance overlap criterion).

Contrast with dissipative chaos.

	Dissipative	**Hamiltonian**
Volume in phase space	Contracts	Preserved
Attractors	Yes (strange attractor)	No
Lyapunov exponents	Sum < 0	Sum $= 0$ (pairs $\pm\lambda$)
Fractal structure	Attractor has $d < n$	Chaotic region has $d = n$
Long-time behavior	Confined to attractor	Diffusive (Arnol'd diffusion)

Example 19.1 (Standard Map)**.** The **Chirikov standard map** is the canonical 2D area-preserving map:

$$p_{n+1} = p_n + K \sin\theta_n, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi}. \tag{86}$$

For $K = 0$: integrable (circles in the (θ, p) torus). For small K: mostly KAM tori with thin chaotic layers near resonances. For $K \gtrsim 0.97$: last KAM torus breaks (Greene's criterion), and global chaos emerges. For large K: widespread chaos throughout phase space.

20 Controlling Chaos

20.1 The OGY Method

Ott, Grebogi, and Yorke (1990) showed that chaos can be **controlled** by exploiting its dense set of unstable periodic orbits. The idea: the chaotic attractor contains infinitely many unstable periodic orbits (UPOs), each embedded in the attractor. By applying small, carefully timed perturbations to a system parameter, one can *stabilize* a chosen UPO.

Algorithm sketch.

1. Choose a target unstable period-n orbit $\{\vec{x}_1^*, \ldots, \vec{x}_n^*\}$ embedded in the chaotic attractor.
2. When the system's trajectory passes near $\vec{x}_k^*$, apply a small parameter perturbation δp such that the trajectory is pushed toward the stable manifold of $\vec{x}_k^*$.
3. Repeat at each return to the Poincaré section.

Because the perturbation is small, the system retains its chaotic nature when control is off. The key advantage over standard stabilization is that no large actuator effort is needed—the chaotic dynamics do most of the work.

Experimental realizations: Magnetic ribbon, cardiac arrhythmias (stabilizing cardiac chaos), laser systems, chemical oscillators.

A Summary of Key Theorems

1. **Existence and Uniqueness** (1.1): Guarantees local existence and uniqueness for C^1 vector fields. Corollary: trajectories cannot cross.
2. **Hartman–Grobman** (5.3): Hyperbolic fixed points of nonlinear systems are topologically equivalent to the corresponding linear system.
3. **Dulac's Criterion** (6.3): Rules out closed orbits via divergence conditions.
4. **Poincaré–Bendixson** (6.4): A confined trajectory in 2D with no fixed points must approach a closed orbit.
5. **Li–Yorke (1975):** Period 3 implies existence of orbits of every period.
6. **Sharkovsky (1964):** Complete ordering of which periods imply which others.
7. **Hopf Bifurcation Theorem:** Eigenvalues crossing the imaginary axis $\Rightarrow$ birth/death of a limit cycle.

8. **Liouville's Theorem (Hamiltonian):** Phase-space volume is preserved; no attractors in Hamiltonian systems.
9. **KAM Theorem:** Most sufficiently irrational tori survive small Hamiltonian perturbations.
10. **Liénard's Theorem:** Conditions for a unique stable limit cycle in Liénard systems.
11. **Singer's Theorem:** Negative Schwarzian derivative $\Rightarrow$ at most one stable orbit per critical point; governs logistic map structure.

B Normal Forms and Center Manifold

B.1 Center Manifold Theorem

When a fixed point has some eigenvalues with zero real part and others with negative real part, the long-time behavior is governed by the **center manifold**.

Theorem B.1 (Center Manifold Theorem—Informal)**.** *Consider $\dot{\vec{x}} = \vec{F}(\vec{x})$ near the origin, with the Jacobian having n_c eigenvalues with $Re = 0$, n_s with $Re < 0$, and n_u with $Re > 0$. Then there exist invariant manifolds:*

- ***Center manifold*** *W^c of dimension n_c: tangent to the eigenspace of the zero-real-part eigenvalues.*
- ***Stable manifold*** *W^s of dimension n_s.*
- ***Unstable manifold*** *W^u of dimension n_u.*

The dynamics on W^c determines the stability and bifurcation type at the fixed point. Trajectories approach W^c exponentially fast (if $n_u = 0$), so the reduced system on W^c captures the essential nonlinear dynamics.

Application to bifurcations. At a saddle-node bifurcation, the Jacobian has one zero eigenvalue. The center manifold is 1-dimensional, and the dynamics on it reduces to the 1D normal form $\dot{x} = r + x^2$—exactly the normal form studied in Section 2.2.

B.2 Normal Forms

Near a bifurcation, coordinate changes can simplify the vector field to a **normal form** that captures the essential dynamics. The key idea: higher-order terms that do not affect the qualitative behavior can be removed by polynomial coordinate transformations.

Bifurcation	**Eigenvalue condition**	**Normal form**
Saddle-node	$\lambda = 0$	$\dot{x} = \mu + x^2$
Transcritical	$\lambda = 0$	$\dot{x} = \mu x - x^2$
Pitchfork (super)	$\lambda = 0$, symmetry	$\dot{x} = \mu x - x^3$
Pitchfork (sub)	$\lambda = 0$, symmetry	$\dot{x} = \mu x + x^3$
Hopf (super)	$\lambda = \pm i\omega$	$\dot{r} = \mu r - r^3$
Hopf (sub)	$\lambda = \pm i\omega$	$\dot{r} = \mu r + r^3$
Period-doubling	multiplier $= -1$	$x_{n+1} = -(1+\mu)x_n + x_n^3$

C Useful Mathematical Facts

C.1 Taylor Expansion for Stability Analysis

For a function $f \in C^n$ near a point x^*:

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2 + \cdots + \frac{f^{(n)}(x^*)}{n!}(x - x^*)^n + O\big((x - x^*)^{n+1}\big).$$

C.2 The Jacobian Matrix

For a vector field $\dot{\vec{x}} = \vec{F}(\vec{x})$ in $\mathbb{R}^n$:

$$J_{ij} = \left.\frac{\partial F_i}{\partial x_j}\right|_{\vec{x}^*}.$$

The eigenvalues of J determine the local stability of the fixed point $\vec{x}^*$.

C.3 Green's Theorem (for Dulac's Criterion)

For a continuously differentiable vector field $\vec{A}(x, y) = (P(x, y), Q(x, y))$ and a simple closed curve C bounding a region D:

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

In divergence form (used for Dulac):

$$\iint_D \nabla \cdot \vec{A}\,dA = \oint_C \vec{A} \cdot \hat{n}\,dl.$$

C.4 Eigenvalues of a 2×2 Matrix

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:

$$\lambda = \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

C.5 Polar Coordinate Conversions

$x = r\cos\theta$, $y = r\sin\theta$; $\quad r = \sqrt{x^2 + y^2}$, $\theta = \arctan(y/x)$.

Useful derivatives:

$$r\dot{r} = x\dot{x} + y\dot{y}, \tag{87}$$

$$r^2\dot{\theta} = x\dot{y} - y\dot{x}. \tag{88}$$